TEMPORAL DYNAMICS OF SOCIAL BALANCE IN FICTIONAL CHARACTER NETWORKS

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Declaration

I hereby confirm that the thesis presented here is my own work, with all assistance acknowledged. I assure that the electronic version is identical in content to the printed version of the Master's thesis.

Hiermit erkläre ich, dass ich die vorliegende Arbeit selbständig verfasst und keine anderen als die angegebenen Quellen und Hilfsmittel verwendet habe. Ich versichere, dass die gedruckte und die elektronische Version der Masterarbeit inhaltlich übereinstimmen.

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Computational resources for LLM-based extraction were provided through the AcademicCloud Chat-AI service (GWDG), without which the corpus would have been considerably smaller and the analysis considerably slower.

AI tool disclosure. LLM assistance was used for language revision and grammar correction during the preparation of this thesis. All intellectual content, analysis, and conclusions are the author's own. The extraction pipeline used Meta Llama-3.3-70B-Instruct exclusively for automated data labeling; this is documented separately in Appendix E and Appendix A.

Abstract

Literary narratives encode complex social systems: characters form alliances, friendships turn into rivalries, conflicts emerge and resolve, and relationships evolve over time in ways that stylize real-world social dynamics. We introduce a dataset of 873 temporal signed character networks, extracted from chapter-level summaries of literary works across multiple genres and historical periods. Characters are represented as nodes and edges encode relationship polarity (positive, negative, or neutral) and their evolution over narrative time, generated via an automated LLM-based extraction pipeline. Applying structural balance theory, we examine recurring motifs and identify genre-specific dynamical signatures. Four principle empirical findings emerge: (1) fictional character networks are universally small-world and densify superlinearly, structural properties that mirror laws observed in real social networks; (2) structural balance, quantified via signed-triad analysis, is persistently high, temporally flat, and genre-stratified, indicating early establishment and maintenance of social structure; (3) temporal motifs reveal dominance of stable alliances $(+++)$ and stable enmities $(---)$, together comprising 74-87% of relationships; and (4) supervised genre classification achieves macro-F1 above stratified baselines, with entropy trajectory features among strongest predictors. This work contributes a large-scale benchmark for computational narrative analysis and demonstrates applicability of network science to cultural systems.

Keywords: temporal networks, signed networks, structural balance, character networks, computational literary studies, narrative analysis, motif analysis, large language models, distant reading

Code and data: https://github.com/L-Earthling/litnet-balance

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Chapter I.

Introduction

1. Motivation and Problem Statement

The opening court of *Hamlet* is, rendered as a static graph, unremarkable: a small, densely connected cluster of positive and neutral ties organized around a royal household. Yet within a single act, the Ghost's revelation converts the Hamlet–Claudius edge from kinship to lethal antagonism, and with it the entire triadic structure of the court becomes unbalanced. Every structural property that makes *Hamlet* legible as tragedy is a property of the *trajectory* of its signed character network, not of any single snapshot.

This observation generalizes. Fictional narratives are bounded, intentionally constructed complex social systems whose meaning resides in change over time. Characters form alliances, betray friends, reconcile with enemies, accumulate or lose social centrality, and reorganize around shifting conflicts in patterns that both reflect and stylize the dynamics of real social worlds. A growing tradition of computational literary studies, initiated by Moretti's call for distant reading [39] and developed through systematic character-network extraction [16, 28, 1, 44], has demonstrated that the relational structure of fiction is amenable to formal network-theoretic analysis. The dominant methodological choice in this literature, however, has been to collapse entire narratives into a single static graph in which an edge records that two characters interacted at some point in the text [28]. Such representations discard precisely the information that distinguishes a tragedy from a comedy, a Bildungsroman from a mystery, or the first act of a play from its last: the sequential reorganization of ties as the story unfolds.

Temporal network science [23, 37] offers the natural formal language for this reorganization, and structural balance theory [11, 21] offers a substantive account of why certain configurations of positive and negative ties are stable

while others are not. Balance theory predicts that social configurations tend toward states in which the "friend of my friend" or "enemy of my enemy" principles are satisfied globally [35]; deviations from balance are transient, driving triadic structures toward one of the two absorbing states — global harmony or two-faction polarization. The combination of temporal signed networks and structural balance has been applied to literary texts only in restricted settings [50, 49], and never at scale across multiple genres and historical periods. The obstacles have been practical: extracting accurate, chapter-resolved, signed relationships from long narratives required manual annotation, and the resulting studies were limited to a handful of canonical works. Recent progress in large language models (LLMs) makes automated extraction of per-chapter signed edges feasible across corpora several orders of magnitude larger than prior work. This thesis exploits that feasibility to address a driving question: *what can temporal signed character networks reveal about the social dynamics and narrative structure of literary texts, and how can evolving character relations be quantified and compared at scale?*

2. Contributions

This thesis makes four contributions.

(i) An end-to-end extraction pipeline. A fully automated pipeline converts chapter-level narrative summaries and curated character lists into temporal signed character networks. Each chapter is processed by an LLM,[1] which identifies character pairs whose relationship is explicitly or implicitly described and assigns a signed polarity label $\sigma \in \{+, -, 0\}$, yielding per-chapter edge tuples (A, B, σ, t). A graph builder then assembles cumulative signed graphs $G_1, \ldots, G_T$ under a *most-recent-wins* convention, so that relationships are modeled as dynamic states rather than cumulative co-occurrence counts: if two characters are allies in chapter 3 but enemies in chapter 7, the edge is coded as negative from chapter 7 onward.

(ii) A dataset of 873 temporal signed character networks. The corpus covers eight genre categories and six historical epochs, ranging from pre-modern drama to twenty-first-century fiction. It comprises 105,550 character-relationship edges across works by 544 distinct authors, with chapter counts ranging from 2 to 162 (median 16) and cast sizes from 2 to 246 (median 16). The dataset — 873 works, 105,550 edges across 544 authors, each represented by per-chapter edge tuples, cumulative graphs, and a 142-dimensional feature vector — is released publicly.

[1] We used `https://huggingface.co/meta-llama/Llama-3.3-70B-Instruct`.

(iii) Empirical analyses of structural and temporal regularities. Using this corpus, the thesis establishes four converging empirical results. Fictional character networks are universally small-world and densify superlinearly, mirroring laws observed in real social networks [31, 23]. The strong-balance index $B(\tau)$ is persistently high and temporally flat across genres, with genre stratification attributable in substantial part to a form confound — Tragedy/Drama is 91.8% plays and Comedy/Satire 66.7% plays, while all remaining genres are predominantly novels — that the thesis explicitly quantifies. Temporal signed motifs reveal genre-specific patterns in how relationships stabilize, destabilize and reconcile. Per-book case studies for representative works from three genres (*Hamlet*, *Pride and Prejudice*, and *Great Expectations*) are provided in Appendix C. Temporal network features carry a modest but statistically reliable genre signal, with entropy trajectory features dominant. An additional finding — a non-monotonic U-shaped epoch effect in $B(\tau)$ — emerges as a co-equal source of structural variation alongside genre.

(iv) Validation of LLM-based extraction against human annotations. The pipeline's extraction quality is evaluated against the publicly available human-annotated character-network data from Massey et al. [36]. Using majority-vote aggregation across annotators as ground truth, the thesis reports entity-level precision, recall and macro-F_1 for character identification and relation-polarity classification. This provides the first quantitative benchmark of LLM-based signed-relation extraction for literary texts and directly addresses a central limitation of prior work, in which the extraction pipeline was not independently benchmarked.

3. Research Questions and Hypotheses

The empirical work is organized around four research questions, each accompanied by an operational test.

RQ1 (Structural regularities) Do fictional character networks exhibit structural regularities documented in empirical social networks, specifically small-world structure and superlinear densification? *Operational test:* compute the small-world coefficient $\sigma_{\text{sw}} = (C_{\text{avg}}/C_{\text{rand}})/(L/L_{\text{rand}})$ [24] and densification exponent α (from $E \propto N^{\alpha}$) for all books with sufficient density. A networks is small-world if $\sigma_{\text{sw}} > 1$ and densifies superlinearly if $\alpha > 1$.

RQ2 (Structural balance trajectories) Does the strong-balance index $B(\tau)$ exhibit genre-specific trajectories consistent with narrative tension con-

ventions? *Operational test:* compute $B(\tau)$ at ten equispaced narrative bins $\tau = t/T$; test genre differences via Kruskal–Wallis with η^2 effect size; and compare the per-book argmin of B to the Freytag climax window $[0.60, 0.75]$ [17]. Two directional hypotheses accompany the omnibus test: *H1 (Freytag alignment)*, that the argmin falls within $[0.60, 0.75]$ for a majority of genres; and *H2 (late-stage resolution)*, that mean balance in the final narrative quintile exceeds mean balance in the middle quintile for genres with conventionally positive resolutions.

RQ3 (Temporal interaction motifs) Are genre-specific patterns detectable in signed temporal interaction motifs extracted from recurring character pairs? *Operational test:* enumerate all length-3 polarity sequences (neutral-excluded) across consecutive appearances of recurring character pairs; compare the resulting eight-type motif frequency distributions across genres via Kruskal–Wallis. An auxiliary hypothesis, *H3 (persistence dominance)*, holds that stable-alliance $(+++)$ and stable-enmity $(---)$ motifs together constitute the dominant share of observations in every genre.

RQ4 (Genre discriminability) Can temporal network features discriminate genre, and which features carry the dominant signal? *Operational test:* train a Random Forest classifier (500 trees, balanced class weights, ten-fold stratified cross-validation) on the 138-dimensional per-book feature matrix; benchmark against majority-class, stratified random, and L1-penalized Logistic Regression baselines; report macro-F_1 and mean decrease in impurity for feature importance. To isolate the form compound, the analysis is repeated on a novels-only subset.

Alongside these four questions, the thesis reports one finding not anticipated at the design stage: the distribution of $B(\tau)$ across six literary epochs is non-monotonic and U-shaped, of magnitude comparable to the genre effect. Because this finding was not predicted *a priori*, it is reported as an exploratory result and interpreted in Section 7 and Section 2.

4. Overview of Results

Table 1.1 summarizes the principal empirical findings; full details are reported in Chapter VII.

Table 1.1.: Principal empirical findings by research question.

RQ	Finding	Key numbers
RQ1	Universal small-world structure	$\sigma_{\mathrm{sw}} > 1$ in 99.3 % of networks; median $\tilde{\sigma}_{\mathrm{sw}} = 2.83$
RQ1	Superlinear densification	97.1 % of books: $\alpha > 1$; median $\tilde{\alpha} = 1.48$
RQ2	$B(\tau)$ persistently high, flat	All genre-mean trajectories in $[0.65, 0.84]$; no systematic monotone trend
RQ2	Freytag climax alignment	$\tilde{\tau}^* = 0.667$; KS $p < 10^{-6}$; 4/8 genres in $[0.60, 0.75]$
RQ2	Genre stratification	$H = 38.8$, $p = 2.1 \times 10^{-6}$, $\eta^2 = 0.062$
RQ3	Stable motif dominance	$+++$ and $---$ together: 76–87 % of all sequences
RQ3	Comedy/Satire distinctive	Highest collapse ($+--$: 0.046) and reconciliation ($--+$: 0.047)
RQ3	Drama distinctive	Only genre: $---$ (0.404) $\approx$ $+++$ (0.389)
RQ4	Temporal features carry signal	RF macro-$F_1 = 0.139$; stratified baseline 0.127; LR reaches 0.196
RQ4	Entropy trajectory dominant	`H_deg_argmin` top predictor (MDI); no $B(\tau)$ features in top 10
Epoch	U-shaped non-monotonic effect	Pre-Modern & 21st-C. highest B; Early 20th-C. lowest; $\Delta B \approx$ 0.10–0.12
Validation	Pipeline benchmarked vs. human	Macro-$F_1 = 0.58$ on 966 pairs / 91 works; within IAA range ($\kappa \approx 0.65$)

5. Thesis Structure

The remainder of the thesis is organized as follows. Chapter II situates the work within computational literary studies, character network research, temporal network science, structural balance theory, and LLM-based information extraction. Chapter III describes the corpus of 873 works, the collection and curation pipeline, the metadata schema, and the form–genre confound that constrains downstream interpretation. Chapter IV details the three-stage extraction pipeline from raw narrative summaries to temporal signed networks. Chapter V defines the network representation and the full set of static, temporal, and balance-theoretic metrics, including the 138-dimensional per-book feature encoding. Chapter VI evaluates extraction quality against the Massey et al. [36] annotations and reports the operational tests for hypotheses H1–H3. Chapter VII presents empirical findings for RQ1–RQ4 and the epoch effect. Chapter VIII interprets these findings, addresses the form–genre confound, and assesses limitations. Chapter IX summarizes contributions and outlines directions for future work, including extension to full primary texts, multilingual comparison, and directed relationship extraction.

Chapter II.

Related Work

This chapter situates the present thesis within five converging bodies of research. Section 1 reviews the tradition of computational literary studies and distant reading that motivates large-scale quantitative analysis of narrative. Section 2 surveys methods for extracting and analyzing character networks from fiction, identifies existing work on signed and temporal extensions, and makes explicit the gap this thesis fills. Section 3 introduces the theoretical apparatus of temporal networks on which our modeling choices rest. Section 4 reviews structural balance theory, from its social-psychological origins to modern quantitative formulations and its sporadic prior application to literary texts. Section 5 covers LLM-based information extraction (IE), the evaluation resources used to benchmark our pipeline, and recent assessments of LLM limitations in literary domains. A consolidated positioning summary closes the chapter in Section 6.

1. Computational Literary Studies and Distant Reading

The application of quantitative methods to literary corpora rests on a methodological premise most prominently articulated by Franco Moretti. In his essay *Conjectures on World Literature* [38], Moretti argued that close reading — the careful, interpretive analysis of individual texts — is structurally incapable of engaging the vast majority of literary production. Literature as a world system is too large for any reader to traverse; meaningful comparative claims require methods that operate at a distance from the individual text. This call for *distant reading* was subsequently operationalized in *Graphs, Maps, Trees* [40], where Moretti demonstrated how quantitative time series, spatial maps and evolutionary diagrams could reveal genre cycles, geographic patterns and morphological constraints that are invisible at the level of individual works. The

anthology *Distant Reading* [39] consolidated these arguments and demonstrated that character interaction graphs compress dramaturgical structure into a form amenable to formal analysis [41].

Matthew Jockers extended this quantitative tradition empirically in *Macroanalysis* [27], applying topic modeling, stylometric analysis and sentiment profiling to a corpus of several thousand 19th-century English-language novels. Jockers showed that computational methods reveal patterns of thematic influence, stylistic periodization and author similarity that no individual reading can expose, and demonstrated the feasibility of drawing literary-historical conclusions from large, machine-readable corpora rather than canonical samples.

Ted Underwood's *Distant Horizons* [53] pursued a closely related agenda with particular rigor regarding historical evidence. Studying trajectories of literary prestige, gender representation and narrative voice across several centuries, Underwood argued that digital methods can adjudicate between competing literary-historical hypotheses and reveal gradual shifts in genre norms that are invisible to scholarship grounded in canonical case studies. His work is notable for attending carefully to sampling bias and for treating computational findings as evidence requiring interpretation rather than as conclusions in themselves. Andrew Piper's *Enumerations* [45] similarly demonstrated that narrative conventions — character types, sentiment distributions, the structure of storyworlds — are not arbitrary but exhibit statistical regularities measurable across thousands of texts, and that these regularities are tied to historical and cultural conditions in traceable ways.

A complementary strand of computational narrative research focuses not on thematic or stylistic aggregates but on the *shape* of narrative time itself. Reagan et al. [48] identified six recurring emotional arc shapes in 1,700 novels; subsequent work extended these arc signatures across media and cultures [8, 43, 51].

2. Character Networks in Fiction

2.1. Foundational Extraction Approaches

The systematic extraction of character networks from literary texts was introduced by Elson, Dames and McKeown [16], who built co-occurrence and conversational-link graphs from 19th-century novels and demonstrated that network topology correlates with literary-critical categories: rural novels are more socially interconnected than urban ones, a structural observation that quantifies an observation that literary criticism had previously only stated

qualitatively. This paper established the core pipeline — character identification, interaction detection, graph construction — that subsequent work has refined. Agarwal et al. [1] extended this approach to Alice's Adventures in Wonderland, demonstrating scale-free degree distributions and a hub-and-spoke topology centered on Alice, exemplifying how network centrality formalizes the literary notion of protagonism. Park et al. [44] examined complex system properties — small-world coefficients, clustering, degree distributions — in networks extracted from both Eastern and Western fiction, providing early empirical evidence that literary social worlds share structural properties with real empirical networks.

Mac Carron and Kenna [33] analyzed networks from the Icelandic sagas and found that while scale-free and small-world properties emerge as in social networks, certain structural anomalies distinguish real from entirely fictional social worlds, suggesting that the social realism of a text leaves measurable structural traces. Bonato et al. [7] subsequently mined character networks from a range of texts and fitted generative models, identifying the parameter regimes that best reproduce observed network statistics. Nalisnick and Baird [42] applied sentiment analysis to extract *signed* character networks from Shakespeare's plays, associating dialogue sentiment with edge polarity — an early instance of the signed extraction paradigm central to the present work.

Labatut and Bost [28] provide the definitive survey of the field, reviewing methods for character identification, coreference resolution, interaction detection, network construction and the full range of downstream analyses across novels, plays, films and TV series. Their taxonomy of edge-construction paradigms and alias-resolution challenges directly informs the pipeline design in Chapter IV.

2.2. Temporal and Dynamic Extensions

Most early work on character networks produced a single static graph aggregating an entire work, discarding the temporal information encoded in the sequential unfolding of chapters, acts or scenes. A smaller body of research has addressed this limitation. Prado et al. [47] modeled literary texts as temporal networks by mapping scenes in the *Iliad* and Norse sagas onto time-ordered interaction events and computing temporal network metrics, demonstrating that the dynamics of character co-appearance carry structural information beyond the static aggregate. Iyyer et al. [26] developed unsupervised methods for tracking *dynamic* character relationships — the trajectory from friendship to rivalry or vice versa — across a corpus of literary texts, showing that relationship polarity evolves in learnable patterns over narrative time. Inoue et al. [25]

applied graph neural network embeddings to full-corpus character networks, constructing low-dimensional representations suitable for downstream tasks such as character similarity.

Most directly related to the present work are two recent studies. Szemes and Nagy [50] introduced explicit temporality into computational drama analysis by computing cumulative network metrics — density, clustering coefficient — act by act across Shakespeare and German drama corpora. They find structural differences between comedy and tragedy that manifest in the developmental trajectory of the network rather than in its final-state snapshot, and demonstrate that creation period does not substantially affect these structural signatures, while genre distinctiveness has declined gradually over literary history. Christou and Tsoumakas [13] study *relational arc dynamics* in a carefully annotated fiction corpus, tracking how character relationship types evolve across narrative time and examining their distribution and diachronic change — providing the closest existing work to the chapter-level temporal signed analysis pursued in this thesis.

2.3. Signed Character Networks

The extraction of signed (positive/negative) character networks from text was formalized by Hassan, Abu-Jbara and Radev [20], who used sentiment classification over textual segments to assign polarity labels to character pairs. Nalisnick and Baird [42] applied this approach to Shakespeare, recovering alliance and antagonism structures that align with literary-critical accounts of the plays. Piper, Xu and Ruths [46] combined citizen science annotation with language models to map the social networks of literary characters at scale, with signed relationship labels derived from a combination of human judgment and model inference — a methodological precedent for the LLM-based extraction pipeline developed in this thesis.

2.4. The Gap This Work Fills

No prior work combines all three: signed edge polarity, chapter-level temporal resolution, and corpus scale sufficient for genre- and epoch-level inference. The present work fills this gap (Section 2).

3. Temporal Networks

3.1. Theory and Representation

Network science traditionally represents systems as static graphs whose edges encode relationships without reference to when or for how long they exist. A growing body of research has shown that this abstraction discards information crucial for understanding dynamics. Holme and Saramäki's foundational review [23] established temporal networks as a distinct subfield, defining time-stamped contact sequences, discussing fundamental concepts such as reachability and transitivity in the temporal domain, and reviewing measures adapted from static network theory — including temporal versions of clustering, path length and centrality. They demonstrated that the temporal ordering of edges fundamentally alters disease spreading, information diffusion and synchronization dynamics relative to time-aggregated predictions, motivating the treatment of time as a first-class feature of network data rather than an annotation to be marginalized.

Masuda and Lambiotte's textbook [37] provides a comprehensive and pedagogically accessible treatment of temporal network theory, covering event-driven representations, interval graphs, aggregation windows and the full suite of metrics used to characterize dynamically evolving networks. Their framework directly informs the representation used in this thesis, in which each literary work is modeled as a sequence of cumulative signed graphs $G_1, \ldots, G_T$ indexed by chapter, with metrics computed at each narrative slice $\tau = t/T$.

3.2. Densification and Structural Growth

A key empirical regularity in growing real-world networks is the *densification law* identified by Leskovec, Kleinberg and Faloutsos [31]. Across social, information and technological networks, the number of edges E grows superlinearly with the number of nodes N: $E \propto N^{\alpha}$ with $\alpha > 1$, meaning each newly added node connects to more than one existing node on average. This densification is accompanied by a *shrinking diameter*: as networks grow, the mean shortest path length decreases rather than increases, contrary to predictions of classical random graph models. In the context of literary networks, the densification law provides a non-obvious benchmark against which fictional social worlds can be compared: if narrative construction is governed by similar organizational principles to genuine social interaction, we should expect superlinear densification as character casts accumulate across chapters.

3.3. Burstiness and Temporal Regularity

Real interaction networks are characterized by *burstiness*: intense activity periods separated by long silences, producing heavy-tailed inter-event time distributions that deviate strongly from Poisson regularity [6]. Goh and Barabási [18] formalized this via a burstiness coefficient $\kappa = (\sigma_\Delta - \mu_\Delta)/(\sigma_\Delta + \mu_\Delta)$, where μ_Δ and σ_Δ are the mean and standard deviation of inter-event intervals; $\kappa \to +1$ indicates maximally bursty, $\kappa = 0$ Poisson regularity, and $\kappa \to -1$ perfectly periodic dynamics. They showed that burstiness and temporal memory are orthogonal dimensions of event-sequence structure, both relevant for characterizing interaction patterns. Applying this measure to character co-appearance streams allows us to quantify whether literary interaction is as sporadic as real social contact or more deliberately structured, a distinction with implications for how faithfully fiction models genuine social dynamics.

3.4. Small-World Structure

The concept of small-world networks — networks with high clustering coefficients but short average path lengths relative to random graphs of equivalent size — was introduced by Watts and Strogatz and has been observed across social, biological and technological networks [54]. Humphries and Gurney [24] established a scalar small-world coefficient $\sigma_{sw} = (C/C_{rand})/(L/L_{rand})$ that normalizes clustering and path length against Erdős–Rényi baselines, allowing direct cross-network comparison. Values $\sigma_{sw} > 1$ indicate small-world structure. Applied to literary character networks, this metric quantifies the degree to which fictional social worlds are more clustered and more efficiently connected than chance, providing evidence about whether authors construct social systems that mirror the structural properties of real social interaction.

4. Structural Balance Theory

4.1. Psychological Origins and Graph-Theoretic Formalization

Structural balance theory (SBT) originates in Fritz Heider's account of interpersonal attitude consistency [21]. Heider argued that cognitive equilibrium is achieved when the sentiments among a triad of individuals satisfy specific sign patterns: a triad of three mutual friends $(+++)$ or a triad with two enemies sharing a common friend or enemy $(+--)$ are *balanced* and psychologically stable. Triads with an odd number of negative edges $(++-$ or $---)$ are *unbalanced* and create cognitive tension that motivates attitude change. Harary

formalized these intuitions within signed graph theory [19], and Cartwright and Harary [11] proved the *structure theorem*: a signed graph is globally balanced if and only if its nodes can be partitioned into two subsets such that all within-subset edges are positive and all between-subset edges are negative — the formal analogue of a world divided into two opposing factions. Easley and Kleinberg [15] provide an authoritative textbook treatment of balance theory in the context of social networks, including its connections to game theory and sociological friendship models.

4.2. Quantification and Partial Balance

For large, dense networks strict global balance is rarely achieved. Aref and Wilson [3] develop a family of *partial balance* measures, including the frustration index — the minimum number of edges that must be removed or sign-flipped to render a network globally balanced — and demonstrate that these measures are more informative than binary balance tests for empirical networks. Their framework accommodates neutral edges and provides continuous measures ranging from zero (maximally imbalanced) to one (globally balanced), directly applicable to signed literary character networks where neutral co-appearances are common. Diaz-Diaz et al. [14] provide a comprehensive review of signed network theory, methods and applications, synthesizing the relationship between balance, spectral properties and dynamical behavior.

4.3. Dynamic Balance and Convergence

Marvel, Kleinberg and Strogatz [35] proved that in a continuous-time model where relationship signs evolve according to balance-promoting dynamics ($dX/dt = X^2$), generic initial conditions lead to exactly two outcomes: either all relationships become positive (global harmony) or the network splits into exactly two hostile factions. This convergence result provides a null model against which temporal balance trajectories in fiction can be interpreted: if narrative social worlds were governed purely by balance dynamics, we would expect monotone convergence toward one of two absorbing states. Deviations from this pattern — persistent partial imbalance, or mid-narrative dips followed by recovery — are thus informative about the distinctive temporal structure of fictional social worlds relative to the dynamics of real signed networks.

4.4. Empirical Signed Networks

Leskovec, Huttenlocher and Kleinberg [30] provided the first large-scale empirical evaluation of SBT using signed networks from online social media platforms (Epinions, Slashdot, Wikipedia elections). They found partial support for balance theory: patterns consistent with SBT's predictions were more common than chance, but an alternative *status theory* better predicted the direction of signed links in directed networks. Their work established the empirical research framework of testing SBT predictions on real network data that the present thesis extends to fictional character networks. Lerner [29] identified a methodological issue in this tradition — that tests of SBT must separate the probability of interaction from the tendency to assign a given sign conditional on interaction — and developed corrected statistical tests showing stronger support for balance in international relations networks.

4.5. SBT Applied to Literary Texts

Prior application of SBT to literary character networks is limited. Nalisnick and Baird [42] computed balance metrics for Shakespeare's plays but did not examine temporal trajectories. Sack [49] applied balance theory to the problem of *narrative generation*, demonstrating that proto-narratives with recognizable dramatic structure emerge from balance-seeking dynamics in simulated signed character networks — a result that suggests balance principles are not merely descriptive but generative of narrative form. These studies are restricted to small corpora and do not examine temporal balance trajectories systematically across genre or epoch.

5. LLM-based Information Extraction

5.1. Large Language Models and Few-Shot Extraction

The transformer-based large language model paradigm, initiated at scale by Brown et al. [10], demonstrated that language models pre-trained on web-scale text corpora can perform IE, relation classification and coreference resolution in a few-shot or zero-shot manner — without task-specific fine-tuning — by conditioning generation on natural-language prompts and examples. This capability makes LLMs attractive for IE in domains where labeled training data is scarce, as is the case for signed character relationship extraction from literary fiction. The subsequent release of open-weight models, notably the LLaMA family [52], extended these capabilities to settings where researchers can run

inference under institutional compute constraints, as in the present project. Liu et al. [32] provide a systematic survey of prompting methods for LLM-based NLP, reviewing prompt design, in-context learning and chain-of-thought reasoning strategies that directly informed the extraction protocol described in Chapter IV. Xu et al. [55] survey LLM-based generative IE specifically, covering entity recognition, relation extraction and event detection, and provide a detailed assessment of accuracy, coverage and calibration tradeoffs that is directly relevant to the reliability concerns addressed in Chapter VI.

The capacity of LLMs for computational social science tasks is reviewed by Ziems et al. [58], who evaluate performance across a range of tasks and find that while LLMs approach human performance on several structured extraction tasks, reliability varies substantially by domain and relationship type. Zhao et al. [56] specifically assessed LLM limitations in literary understanding, finding that current models systematically miss implicit relationship cues in detective fiction — a finding relevant to interpreting the recall limitations of our pipeline, particularly for indirect or ironic negative relationships.

5.2. Coreference, Entity Linking and Long-Document Processing

A fundamental challenge for character network extraction from book-length texts is coreference resolution: the same character may be referred to by proper name, pronoun, title, nickname and narrative description across hundreds of thousands of tokens. Bamman et al. developed models for character persona inference [5] and BookNLP [4], an end-to-end NLP pipeline for book-length texts that has become a standard tool in computational literary studies.

Martinelli et al. [34] established BOOKCOREF, the first book-scale coreference benchmark (200k+ tokens), demonstrating performance drops of up to 20 CoNLL-F1 points at full-book scale. Dual-LLM and markdown-annotation approaches improve performance in chunked long narratives [57, 22]. Taken together, these results establish that LLM-based approaches provide a viable path for character identity tracking in chapter-level summaries — the methodological setting of this thesis.

Amalvy, Labatut and Dufour [2] investigated the downstream impact of NER and coreference quality on character network construction, finding that NER performance is text-dependent and strongly affects character detection, that NER alone misses many co-occurrences and that coreference resolution is necessary to recover them. They also find that LLM end-to-end pipelines

are outperformed by traditional NLP pipelines in recall, contextualizing the extraction quality analysis in Chapter VI.

5.3. Annotation Resources for Evaluation

Massey et al. [36] produced a human-annotated dataset of character relationships in literary fiction, labeling character pairs with relationship types and polarities across a set of novels drawn from multiple genres. This resource constitutes the primary external benchmark against which the LLM extraction pipeline in this thesis is validated: precision, recall and F1 for signed relationship labels are computed by comparing automatically extracted edges against. Massey et al.'s annotations using majority-vote aggregation across annotators. Christou and Tsoumakas [12] subsequently released the ARF (Artificial Relationships in Fiction) dataset, a synthetically annotated relation extraction resource built from Project Gutenberg fiction using GPT-4o with a fiction-specific ontology, providing a complementary resource for evaluating entity and relation extraction models.

6. Summary and Positioning

The foregoing review identifies a clear gap at the intersection of five active research traditions. Computational literary studies has established that large-scale quantitative analysis of narrative is both legitimate and productive, but most corpus-scale work operates on scalar features (sentiment, topics, style) rather than relational network structure. Character network research has moved from static extraction to dynamic modeling and from unsigned to signed graphs, but these advances have not been combined at corpus scale: signed networks appear in small annotated studies; temporal analysis appears in dramatic corpora but without signed polarity; large-scale extraction remains predominantly unsigned and static. Temporal network science provides a rigorous analytical framework — densification, burstiness, small-world structure, cumulative graph sequences — that has not been systematically applied to literary character networks. Structural balance theory offers a principled account of relationship stability and tension that has been tested on social media networks and simulated dramatic texts but never at scale across a multi-genre literary corpus. And LLM-based information extraction has made signed relationship labeling feasible in low-resource domains, but extraction pipelines for literary fiction have not been benchmarked against human annotation at book scale.

This thesis addresses all five gaps simultaneously. It contributes (i) an LLM-based extraction pipeline that scales to 873 literary works across eight genres and six historical epochs; (ii) a dataset of temporal signed character networks with chapter-level resolution validated against a human-annotated benchmark; (iii) a systematic analysis of structural regularities — small-world structure, superlinear densification, burstiness, edge persistence — across the full corpus; (iv) the first large-scale empirical test of structural balance theory in literary networks, revealing persistent high balance, genre stratification, and a non-monotonic epoch effect of comparable magnitude; and (v) a genre classification study establishing that temporal features carry reliable genre signal with entropy trajectory as the dominant discriminative dimension. Taken together, these contributions provide both a methodological foundation and a set of empirical findings that bridge computational literary studies and network science in a way not previously achieved in the literature.

Chapter III.

Data and Corpus

1. Data Collection

The corpus is constructed from publicly available chapter-level narrative summaries and character lists provided by online literary study guides. For each work, two structured plain-text files are collected: a summary file containing *per-chapter summary narrative text* delimited by standardized partition markers, and a *character file* enumerating the cast with a major/minor distinction and brief descriptive annotations.

Data collection was performed via an automated pipeline using HTTP requests and HTML parsing. The parsed content was saved to a fixed directory structure that mirrors the subsequent processing stages, ensuring reproducibility of corpus construction independent of LLM inference.

Chapter-level summaries rather than primary texts serve as the empirical basis for a practical reason: summaries are consistently structured, segmented into discrete narrative units, and available at scale. For a corpus of this size, this enables systematic comparison across hundreds of works without the pre-processing overhead — irregular chapter boundaries, OCR artefacts, and highly variable text length — that full-text corpora impose. The key limitation is *compression*: study-guide summaries simplify and abbreviate relative to primary texts, and secondary or episodic characters may be underrepresented or omitted. These constraints are acknowledged throughout and are addressed in the discussion of extraction validity (Section 4).

2. Corpus Curation and Filtering

Filtering pipeline. The initial collection pool comprised approximately 900 candidate works drawn from two complementary study-guide platforms, com-

bined after systematic deduplication. A two-pass matching procedure was applied to identify and remove overlapping titles: exact title matching was followed by sub-string and author co-occurrence checks, removing 168 duplicate entries. Non-fiction works were excluded using keyword-based filtering on genre metadata, flagging titles categorized as memoir, autobiography, essay, philosophy, or history. The filtering further excluded poetry without a sustained narrative character structure, anthologies, and multi-volume compiled editions that lack a unified cast.

Inclusion criteria required works to be (i) English-language fiction, (ii) attributed to a single author and constituting a single work, and (iii) segmented into at least two narrative partitions. Works where the automated extraction pipeline produced no valid network output, owing to extraction failures or networks with fewer than two characters and one ed, were removed at a post-processing stage.

A secondary quality-control pass targeted structural anomalies in the extracted networks. Works exhibiting both large cast size ($N > 50$) and near-zero density ($\delta < 0.03$) were individually inspected: this combination indicates independent sub-stories with no shared characters, or a philosophical treatise misclassified as fiction. Works of this type were removed. Full inclusion and exclusion criteria are listed in Appendix B. Conversely, epic poems with sustained cross-partition character interaction, including *The Divine Comedy*, *The Aeneid*, and the *Mahabharata*, were retained despite their verse form, as they exhibit recurrent character interaction across partitions.

The final analysis corpus comprises **873 works**.

Genre taxonomy. Genre labels were derived from the source platforms' categorization metadata and consolidated into eight primary categories (Table 3.1). The taxonomy was designed to maintain $n \geq 30$ works per genre, a threshold chosen to ensure that per-genre statistical estimates are based on adequate sample sizes for the Kruskal–Wallis tests and classifier evaluations reported in subsequent chapters. Approximately one-sixth of works (147 of 873) could not be assigned a reliable primary genre owing ambiguous multi-genre categorization; these works are included in all aggregate structural analyses but excluded from genre-stratified comparisons.

Form–genre confound. A critical structural confound is present in the corpus that is flagged here and revisited throughout: *Tragedy/Drama* is 91.8% plays (78 of 85 works); *Comedy/Satire* is 66.7% plays (32 of 48). All other genres are predominantly novels. Because plays typically feature smaller, more densely

interconnected casts than novel summaries, any observed structural difference between these two genres and the remaining six must be interpreted with caution: it may reflect a form effect rather than — or in addition to — a genre-content effect. Form-stratified robustness analyses are provided where relevant (see Section 3).

3. Corpus Overview and Statistics

Table 3.1 summarizes the 873-work analysis corpus by primary genre, literary form, and historical epoch. Of the 873 works, 726 carry a primary genre label and are used in genre-stratified analyses; the remaining 147 are included only in aggregate comparisons. The corpus spans 544 distinct authors and yields a total of 105,550 character-relationship edge observations across all extracted networks. Chapter counts range from 2 to 162 (median 16); cast sizes from 2 to 246 (median 16). The large cast-size maximum reflects a small number of works with highly populous or episodic character structures that survived post-processing filters; the median is a more representative indicator of typical network scale.

The genre distribution is markedly skewed: *Literary / Realist Fiction* accounts for 39% of genre-labeled works (286 of 726), reflecting both the category's broad scope — it encompasses 19th-century realism, modernist fiction, and contemporary literary novels — and its prevalence in the source material. The next largest categories, *Coming-of-Age / Bildungsroman* (107) and *Tragedy / Drama* (85), each exceed the $n = 30$ threshold by a substantial margin, while *Romance / Domestic Fiction* (36) sits closest to that floor.

Epoch coverage is concentrated in the 20th and 21st centuries: taken together, the three post-1900 cohorts account for approximately 76% of the corpus. The Pre-Modern and Early Modern strata are correspondingly small (roughly 6% each). This distribution reflects the composition of the source study-guide platforms rather than a deliberate sampling strategy and must be borne in mind when interpreting epoch-stratified analyses: estimates for the pre-1600 stratum in particular are based on small n and carry wider uncertainty.

The form–genre cross-tabulation in Table 3.1 makes the confound concrete: genres outside Tragedy/Drama and Comedy/Satire are novel-dominant ($> 90\,\%$), so structural comparisons involving the two play-dominated genres must be interpreted with caution throughout (Section 3).

Table 3.1.: Analysis corpus ($n = 873$) by primary genre, literary form, and historical epoch. Genre-level form counts cover the 726 works with a primary genre label. Tragedy/Drama and Comedy/Satire are play-dominated — a form–genre confound addressed throughout this thesis (see Section 3). Epoch counts are drawn from the full metadata pool ($n = 900$); the 27-work gap relative to the analysis corpus reflects post-processing exclusions and does not materially affect epoch percentages.[a]

Genre	n	**Novel**	**Play**	**Other**
Literary / Realist Fiction	286	253	4	29
Coming-of-Age / Bildungsroman	107	98	2	7
Tragedy / Drama	85	7	78	0
Fantasy / Adventure	65	58	1	6
Science Fiction / Dystopia	52	50	0	2
Comedy / Satire	48	15	32	1
Mystery / Thriller	47	44	0	3
Romance / Domestic Fiction	36	34	0	2
Subtotal (genre-labelled)	726	559	117	50
No genre label	147	–	–	–
Total	**873**			

Epoch	n[a]	%[a]
Pre-Modern (pre-1600)	51	5.7
Early Modern (1600–1800)	52	5.8
19th Century	112	12.4
Early 20th Century (1900–1950)	158	17.6
Late 20th Century (1950–2000)	311	34.6
21st Century (2000+)	215	23.9
Total	**900**[a]	100.0

[a] Epoch counts are from the 900-entry metadata pool. The 27-book difference relative to the 873-book analysis corpus reflects works removed during post-processing; the proportional distribution is not materially affected.

4. Metadata Schema

Table 3.2 documents the eight fields of the metadata schema released with the dataset.

Table 3.2.: Metadata schema fields for the combined corpus.

Field	Description
`Author`	Author name (mixed case).
`Book`	Work title.
`Year`	Publication year (integer).
`Epoch`	Historical epoch label (one of six categories; see below).
`Genre_Primary`	Primary genre label (one of eight categories; absent for 147 works).
`Genre_Secondary`	Secondary genre label where applicable; empty for works assigned a single genre.
`Form`	Literary form: `Novel`, `Play`, or `Other`.
`Remarks`	Free-text notes (e.g. non-fiction flag, short-story collection, canonical translation).

Genre labels consolidate source-platform categorizations into eight primary classes; approximately 35 % of entries carry a secondary label, retained in `Genre_Secondary`. Works with absent or contradictory metadata are left unlabeled ($n = 147$). Epoch boundaries follow established literary-historical

Chapter IV.

Extraction Pipeline

1. Pipeline Overview

The construction of temporal signed character networks from chapter-level summaries proceeds through three sequential stages (Figure 4.1). Stage 1 collects and structures the textual source material (per-chapter narrative summaries and curated character lists) into a standardized input format; the corpus and its preparation are described in Chapter III. Stage 2 processes each chapter partition in turn with an LLM, yielding a tabular record of signed edge tuples. Stage 3 accumulates these tuples into a sequence of cumulative signed graphs constituting the temporal network representation of each literary work. The output (one network CSV file per book) forms the primary dataset for all analyses reported in Chapter VII.

The pipeline is implemented in Python 3.12; all configuration parameters (model version, temperature, prompt text, and rate-limit settings) are stored in a single configuration cell and written to a run log for reproducibility. A SQLite progress database records completion status at partition granularity.

2. Text Preprocessing and Partitioning

Each book's source material is stored as two plain-text files in a dedicated directory: a summary file (`<book>_clean.txt`) containing the per-chapter narrative text, and a character file (`<book>_characters_clean.txt`) containing the curated character list.

Partition format. The summary file is divided into partitions using the fixed delimiter:

```
=== PARTITION N: [Chapter Title] ===
```

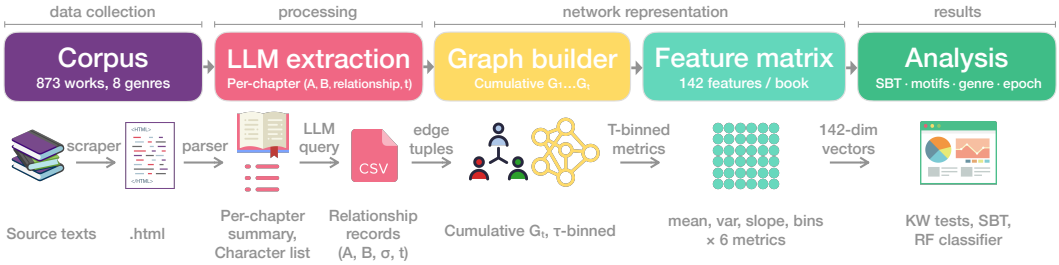

Figure 4.1.: End-to-end pipeline. (1) Chapter summaries and character lists are collected from publicly available sources. (2) LLM extracts signed character-pair edge tuples (A, B, σ, t) per chapter. (3) Graph builder assembles cumulative temporal networks $G_1, \ldots, G_T$. (4) Per-book feature vectors feed the (5) analyses reported in Chapter VII.

where N is the integer chapter index and *Chapter Title* is the descriptive heading (e.g. "Chapter One: The Shire" or "Act II, Scene 3"). The delimiter is matched by the regular expression `===\s*PARTITION\s+(\d+)\s*:\s*(.+?)\s*===` (case-insensitive), extracting both the index and title as captured groups. The parser splits the file at each match boundary and assigns the intervening text as the partition content; partitions with empty content after stripping whitespace are discarded.

Character list format. The character file is structured into two named sections, `MAJOR CHARACTERS` and `MINOR CHARACTERS`, with one entry per line: the character's canonical name followed by a brief description in parentheses. This format supplies the LLM with a closed set of canonical identifiers, reducing alias fragmentation in the model output (character variants such as "Lizzy", "Miss Bennet", and "Elizabeth" are consolidated to a single node label by instruction). The MAJOR/MINOR distinction communicates relative narrative importance, guiding extraction decisions for peripheral characters that appear only briefly in a given chapter. Character files were prepared manually to ensure name accuracy and cross-partition consistency.

3. LLM-based Relationship Extraction

3.1. Model

Relationship extraction is performed by Meta's *Llama-3.3-70B-Instruct*[1] [52], a publicly available 70B-parameter instruction-tuned model. The model was accessed via the KISSKI Chat-AI service of AcademicCloud,[2] using an OpenAI-compatible chat completions endpoint. All calls use `temperature` = 0.1, which minimizes stochastic variation while retaining the model's capacity to handle ambiguous relational signals, and `max_tokens` = 1500, sufficient to accommodate JSON output for the largest casts in the corpus. API configuration details and wall-clock time estimates are reported in Appendix F.

3.2. Prompt Structure

For each chapter partition, a prompt is assembled from four components submitted as a single user message (full template in Appendix A).

Character context. The full character list constrains the model to use names corresponding directly to graph nodes, reducing alias fragmentation.

Partition content. The chapter summary text provides the primary evidential basis for extraction.

History context. Up to 30 relationship records from the three most recently processed partitions are appended, providing continuity cues for relationships not explicitly mentioned in the current chapter. The 30-row cap prevents history from displacing current partition content.

Output specification. The model is instructed to return a JSON array of tuples $(chapter, char_a, char_b, relationship)$ with $relationship \in \{$`positive`, `negative`, `neutral`$\}$. Markdown code fences are stripped before parsing; records with values outside this set are discarded.

4. Infrastructure

Three mechanisms ensure zero data loss on interruption. (i) A SQLite progress database records completion status at partition granularity; completed partitions are skipped on restart. (ii) Extracted relationships are appended to the

[1] https://huggingface.co/meta-llama/Llama-3.3-70B-Instruct

[2] Operated by GWDG (Gesellschaft für wissenschaftliche Datenverarbeitung mbH Göttingen) https://academiccloud.de/services/chatai/

per-book output CSV after each partition. (iii) A `RateLimitedClient` wrapper enforces 13 requests per minute per key; a `MultiKeyCoordinator` routes calls across available keys in round-robin order, scaling throughput linearly with the number of active keys.

5. Temporal Network Construction

5.1. Cumulative Signed Graphs

The per-book extraction output is a flat CSV with columns `Book`, `Chapter`, `Character A`, `Character B`, `Relationship`. Each row is a signed edge tuple (A, B, σ, t) where $\sigma \in \{+, -, 0\}$ and t is the chapter index. From these tuples the graph builder assembles a sequence of cumulative signed graphs $G_1, \ldots, G_T$, where

$$G_t = (V_t,\, E_t,\, \sigma_t),$$

V_t is the set of characters observed through chapter t, E_t is the set of character pairs with at least one extracted relationship through chapter t, and $\sigma_t : E_t \to \{+, -, 0\}$ records the most recently observed polarity for each edge.

Most-recent-wins convention. For a pair (A, B) observed with polarity σ' at chapter $t' > t$:

$$\sigma_{t'}(A, B) = \sigma', \quad \forall\, t \geq t'.$$

This models relationships as dynamic states: if two characters are allies in chapter 3 but antagonists by chapter 7, their edge is coded as negative from chapter 7 onward.

Normalized narrative time. To enable cross-book comparison, the chapter index is normalized to $\tau = t/T \in (0, 1]$ (5.1), where T is the total chapter count. Under this normalization $\tau = 0.5$ always corresponds to the narrative midpoint regardless of absolute chapter count; τ-binned snapshots at ten equispaced positions $\tau \in \{0.05, 0.15, \ldots, 0.95\}$ yield fixed-length temporal feature representations comparable across books. The per-book feature vectors derived from these snapshots are described in Chapter V.

5.2. Undirected Construction and Edge Polarity Handling

All graphs are undirected by construction: the extraction prompt elicits relationship descriptions without anchoring to either character's perspective.

Duplicate pairs (A, B) and (B, A) are eliminated by canonical lexicographic ordering.

Neutral edges ($\sigma = 0$) are retained in the graph and contribute to all connectivity-based metrics (cast size, edge count, density, degree centrality, community structure, cohesion). They are excluded from the triadic census used to compute the structural balance index $B(\tau)$: the Cartwright–Harary balance criterion [11, 21] applies only to signed (positive or negative) edges, and triads containing a neutral edge have no determinate balance classification under strong-balance theory. This treatment keeps the connectivity representation maximally inclusive while ensuring that balance statistics are computed over a well-defined subset of edges.

The resulting per-book CSV files from both sub-corpora are concatenated into a single flat table, from which the per-book feature matrices used in Chapter VII are derived.

Chapter V.

Network Representation and Metrics

This chapter formalizes the network representation underlying all analyses reported in Chapter VII and defines the complete set of metrics computed from it. Section 1 specifies the cumulative signed graph model and the normalization of narrative time. Sections 2 and 3 define static and temporal structural metrics respectively. Section 4 details the structural balance operationalization, including the triad census and the temporal motif scheme. Section 5 describes the feature vector construction and classifier setup used for genre prediction.

1. Temporal Network Representation

Definition V.1 (Cumulative signed graph):
At narrative step $t \in \{1, \ldots, T\}$, the graph $G_t = (V_t, E_t, \sigma_t)$ is defined as follows:

- V_t is the set of all characters mentioned in partitions $1, \ldots, t$.
- $E_t \subseteq \binom{V_t}{2}$ is the set of all unordered character pairs $\{A, B\}$ for which at least one relationship was extracted in partitions $1, \ldots, t$.
- $\sigma_t : E_t \to \{+1, 0, -1\}$ assigns to each edge the polarity of its most recently observed extraction: $\sigma_t(\{A, B\}) = \sigma_{t'}(\{A, B\})$, where $t' = \max\{s \leq t : (A, B, \cdot, s) \text{ was extracted}\}$.

The *most-recent-wins* convention treats relationships as dynamic states rather than aggregates. If characters A and B are allies in partition 3 but enemies in partition 7, their edge carries a negative sign from partition 7 onward. This models the narrative intuition that a recent betrayal overrides an earlier alliance. All graphs in the sequence $G_1, G_2, \ldots, G_T$ are undirected; the extraction pipeline (Chapter IV) produces symmetric dyads, and no reciprocal $(A, B) / (B, A)$ pairs are present in any E_t.

Narrative time is normalized to

$$\tau = \frac{t}{T}, \qquad t = 1, \ldots, T, \tag{5.1}$$

so that $\tau \in (0, 1]$ and $\tau = 0.5$ always corresponds to the narrative midpoint, regardless of chapter count. This mapping permits direct comparison across works of vastly different length — from two-chapter novellas to 162-chapter novels in the present corpus — by aligning all trajectories on a common scale.

The terminal graph G_T records the full cumulative network at the end of the narrative and serves as the basis for all static metrics (Section 2). Temporal metrics (Section 3) track how structural properties change along the sequence $G_1, \ldots, G_T$.

2. Static Network Metrics

Static metrics are computed on the terminal graph G_T.

Cast size and edge count: $N = |V_T|$ denotes the total number of distinct characters; $E = |E_T|$ the total number of unique character pairs. Across the full corpus of 873 works, the median cast is $\tilde{N} = 16$ characters and the median edge count is $\tilde{E} = 29$.

Density:

$$\delta = \frac{2E}{N(N-1)}, \tag{5.2}$$

the fraction of possible undirected pairs that are realized. The mean density across the corpus is $\bar{\delta} = 0.304$ ($\sigma = 0.197$).

Clustering coefficient: $C = \overline{C_i}$, the mean local clustering coefficient over all nodes i with degree ≥ 2, quantifies the tendency of character neighborhoods to form closed triangles (triadic closure). The corpus mean is $\bar{C} = 0.538$.

Shannon degree entropy:

$$H = -\sum_i p_i \log_2 p_i, \qquad p_i = \frac{\deg(i)}{\sum_j \deg(j)}, \tag{5.3}$$

where the sum runs over all characters in V_T. H measures how evenly connectivity is distributed across the cast: high H indicates a diffuse, ensemble-type

social world, while low H reflects a strongly hub-dominated network with one or two characters concentrating most ties. This metric proves to be the dominant genre-discriminative signal (Section 6).

Small-world coefficient:

$$\sigma_{\text{sw}} = \frac{C_{\text{avg}}/C_{\text{rand}}}{L/L_{\text{rand}}}, \tag{5.4}$$

where C_{rand} and L_{rand} are the mean clustering coefficient and mean shortest path length of 100 Erdős–Rényi random graphs with the same N and E [24]. $\sigma_{\text{sw}} > 1$ indicates that the observed network is simultaneously more clustered and more efficiently connected than the corresponding random baseline — the defining property of small-world structure [54]. The coefficient is computed for the 696 books whose terminal graphs have sufficient density for path-length estimation; 177 graphs were too sparse (mean path length undefined due to disconnected components).

Edge polarity fractions: $f^+ = n_+/E$, $f^- = n_-/E$, and $f^0 = n_0/E$, where n_+, n_-, and n_0 count positive, negative, and neutral edges in G_T. These fractions capture the aggregate social tone of each work and constitute the single strongest static genre discriminator in Kruskal–Wallis testing (Table 7.1).

3. Temporal Network Metrics

Density trajectory: $D(\tau)$ denotes the graph density (5.2) evaluated at narrative position τ, averaged across books within a genre or across the full corpus. The trajectory captures how the fraction of realized character pairs changes as new characters are introduced.

Signed density components: $D^+(\tau) = n_+(\tau)/\binom{|V_t|}{2}$ and $D^-(\tau) = n_-(\tau)/\binom{|V_t|}{2}$ measure the density of positive and negative edges respectively at narrative position τ. The ratio $R^+(\tau) = n_+(\tau)/E_t$ tracks the fraction of edges carrying positive polarity at each step.

Character introduction rate: $\delta_N(\tau) = |V_t|/|V_T|$ records the cumulative fraction of the final cast that has been introduced by narrative position τ, tracing the rate at which new characters enter the social world.

Edge persistence: For each character pair, the *persistence* is defined as the mean length of consecutive-partition co-appearance runs — the average number of successive chapters in which the pair appears without interruption. Higher persistence indicates that once established, relationships are maintained across many consecutive narrative units. Persistence is computed from the raw event stream extracted in Chapter IV.

Inter-appearance burstiness: The burstiness of a character pair's co-appearance pattern follows [18]:

$$\kappa = \frac{\sigma_\Delta - \mu_\Delta}{\sigma_\Delta + \mu_\Delta}, \tag{5.5}$$

where μ_Δ and σ_Δ are the mean and standard deviation of the inter-appearance intervals (in chapters) for a given pair. $\kappa = -1$ indicates perfectly regular appearances; $\kappa = 0$ corresponds to a Poisson process; $\kappa > 0$ indicates bursty, heavy-tailed behavior. Across the full corpus, all genre–polarity combinations yield $\kappa \in [-0.45, -0.71]$, uniformly negative, indicating that character co-appearances in fiction are more regular than a Poisson process — the opposite of the bursty patterns documented in real-world interaction networks [6]. This regularity reflects deliberate authorial construction rather than the self-organized dynamics of genuine social interaction.

Protagonist centrality trajectory: The protagonist proxy is the character with the highest median betweenness centrality across all narrative partitions. Its normalized degree centrality $\hat{d}_{\text{prot}}(\tau) = \deg_t(\text{prot})/(|V_t| - 1)$ is tracked across τ to characterize hub dominance and how it evolves over the narrative. Centrality rank stability across partitions is quantified by the median Spearman rank correlation ρ between consecutive-chapter centrality rankings.

Edge volatility: The fraction of edges that change polarity between consecutive partitions, averaged across all transitions in a book. Across the corpus, the mean edge volatility is 37.4%, indicating that most relationships persist in sign from one chapter to the next.

Densification exponent: Fitting

$$E \propto N^\alpha \tag{5.6}$$

per book via ordinary least squares on $(\log N, \log E)$ pairs across the cumulative snapshot sequence quantifies how rapidly edges accumulate relative to cast

growth. $\alpha > 1$ indicates superlinear densification: each newly introduced character forms, on average, more than one relationship with the existing cast. The corpus-wide median is $\tilde{\alpha} = 1.48$ and 97.1 % of books yield $\alpha > 1$ (Section 2.2).

4. Structural Balance Analysis

Strong-balance index: Structural balance theory [21, 11] classifies triads of characters according to the product of their three edge signs. Under the Cartwright–Harary strong-balance criterion, a closed signed triad $\{i, j, k\}$ is *balanced* if the product $\sigma(\{i, j\}) \cdot \sigma(\{j, k\}) \cdot \sigma(\{i, k\})$ is positive — equivalently, if the triad is of type $(+++)$ (mutual alliance) or $(+--)$ (two allies sharing a common enemy). Triads of type $(++-)$ or $(---)$ are *unbalanced*. The strong-balance index at narrative position τ is:

$$B(\tau) = \frac{|\{\text{closed signed triads at } \tau \text{ that are balanced}\}|}{|\{\text{closed signed triads at } \tau\}|}. \tag{5.7}$$

$B(\tau) = 1$ corresponds to a globally balanced network; $B = 0.5$ is the expectation for a randomly signed graph. Neutral edges are excluded from the triad census throughout: only triads whose three edges all carry a strictly positive or negative polarity are counted, since a neutral edge does not contribute directionally to social tension or cohesion.

Minimum triad threshold: Balance estimates derived from very few triads are subject to high variance that renders genre comparisons unreliable. A minimum of five signed triads per narrative slice is therefore imposed; slices below this threshold are excluded from $B(\tau)$ computation. This threshold was determined empirically as the minimum at which per-genre mean trajectories stabilize across bootstrap resamples of the corpus. Applying it excludes 268 books — predominantly short works with sparse signed interactions — leaving $n_{\text{valid}} = 605$ books with valid $B(\tau)$ data for the trajectory analyses in Chapter VII.

Temporal interaction motifs: A *temporal interaction motif* of length 3 is defined as a sequence of three consecutive appearances of the same character pair, with the polarity at each appearance recorded. Neutral-edge appearances are excluded, so each appearance contributes a sign from $\{+, -\}$. Eight motif types are therefore possible: $(+++)$, $(---)$, $(+-+)$, $(-+-)$, $(++-)$, $(+--)$, $(--+)$, $(-++)$. Stable-persistence motifs — $(+++)$ (stable alliance)

and $(---)$ (stable enmity) — capture relationships that maintain their sign across three consecutive appearances. Alternating motifs $(+-+)$ and $(-+-)$ capture rapidly oscillating relationships. Transition motifs $(++-)$, $(+--)$, $(--+)$, $(-++)$ capture one-directional sign changes. Frequencies are normalized within genre by the total number of valid length-3 sequences per book, enabling cross-genre comparison that is not confounded by corpus size. Neutral exclusion is motivated by the same consideration as in the triad census: neutral edges contribute no directional information about relational stability or destabilization.

5. Genre Classification Feature Vector

Feature vector construction: For each of the 873 books, a 138-dimensional numeric feature vector is assembled from the temporal metrics defined in Sections 2–4. The vector encodes the trajectory shape of seven metrics — cumulative density D, positive signed density D^+, negative signed density D^-, positive edge ratio R^+, Shannon degree entropy H, clustering coefficient C, and structural balance index B — using a fixed set of 19 features per metric: values at ten equispaced τ-bins $\{0.05, 0.15, \ldots, 0.95\}$, together with nine summary statistics (mean, linear slope, variance, τ-position of minimum, τ-position of maximum, value at the first and last narrative partition, end-minus-start change δ_m, and range). This yields $7 \times 19 = 133$ trajectory features. Nine scalar features are appended: the densification exponent α, overall edge volatility, the narrative half-life τ_{50} (the position at which 50 % of the final cast has been introduced), the τ-position of minimum balance τ_B^*, and the end-minus-start balance change δ_B. Missing values arising from books with insufficient triads for $B(\tau)$ or from disconnected graphs are imputed with column-wise medians prior to classification.

Classifier: A Random Forest [9] with 500 trees, balanced class weights (to correct for class imbalance), and 10-fold stratified cross-validation serves as the primary classifier. Performance is reported as macro-averaged F1 across all eight genre classes, which weights each class equally regardless of corpus frequency. Feature importance is measured by mean decrease in impurity (Gini importance), averaged across all 500 trees.

Baselines: Three baselines calibrate the interpretation of classifier performance:

(i) *Majority-class dummy*: always predicts Literary / Realist Fiction, the most frequent class ($n = 286$). This establishes the uninformative floor (macro-F1 = 0.071).

(ii) *Stratified-random dummy*: samples predictions according to observed class proportions, providing an imbalance-corrected chance level (macro-F1 = 0.127).

(iii) *L1-penalized Logistic Regression*: a linear separability ceiling that establishes how much signal can be extracted without non-linear interactions (macro-F1 = 0.196).

The majority-class dummy reveals the degree of class imbalance; exceeding it confirms only that some discriminative signal is present. Exceeding the stratified baseline is a more demanding criterion: it establishes that temporal network features carry genre information above what chance assignment could achieve given the class distribution. A novels-only robustness analysis ($n = 537$, Tragedy/Drama and Comedy/Satire excluded) controls for the form–genre confound documented in Chapter III: because Tragedy/Drama is 96 % plays and Comedy/Satire is 71 % plays, structural differences attributable to dramatic form rather than genre content could inflate apparent genre discriminability in the full-corpus result. A pairwise discriminability matrix (binary Random Forest, 5-fold cross-validation, one classifier per genre pair) further reveals which specific genre pairs are structurally distinguishable and which occupy overlapping regions of feature space.

Chapter VI.

Validation

1. Extraction Quality: Comparison with Human Annotations

1.1. Gold Standard

Massey et al. [36] provide character relationship annotations for 109 literary works. Each work is annotated by at least two independent annotators, who assign a polarity label (+, −, or neutral) to each observed character pair based on the holistic relational character of the work. The authors report inter-annotator agreement of approximately $\kappa \approx 0.65$ (Cohen's κ), indicating substantial but imperfect agreement — a figure that reflects the genuine interpretive ambiguity of polarity classification in literary contexts. Of the 109 annotated works, 91 appear in our corpus, yielding 966 matchable character pairs after deduplication and canonical normalization.

1.2. Matching Methodology

Massey et al. [36] annotate relationships at book level (a single holistic label per pair), whereas our pipeline produces chapter-level labels that evolve over narrative time. To bridge this difference, we aggregate our chapter-level labels to a book-level *majority-vote* label per canonical character pair:

$$\hat{\sigma}(A, B) = \arg\max_{\sigma \in \{+,-,0\}} \sum_{t=1}^{T} \mathbf{1}[\sigma_t(\{A, B\}) = \sigma]. \tag{6.1}$$

Character pairs are matched using a normalized canonical form that strips articles, honorifics, and punctuation, and treats (A, B) and (B, A) as the same

pair. Pairs that cannot be matched after normalization are excluded; these constitute 258 unmatched pairs.

1.3. Results

Table 6.1 reports precision, recall, and F_1 per polarity class together with macro-averaged figures.

Table 6.1.: Extraction quality benchmarked against [36] human annotations. 966 matched character pairs across 91 works. Macro-F_1 is the headline metric.

Class	**Precision**	**Recall**	F_1	**Support**
Positive	0.73	0.76	0.74	516
Negative	0.66	0.70	0.68	277
Neutral	0.35	0.29	0.32	172
Macro avg	0.58	0.58	**0.58**	965

The full per-class breakdown and pairwise discriminability matrix are reported in Appendix D (Tables C.2 and C.3).

The pipeline achieves macro-$F_1 = 0.58$, with strong recovery of positive ($F_1 = 0.74$) and negative ($F_1 = 0.68$) relationships. The neutral class is substantially harder ($F_1 = 0.32$), a pattern that is expected and interpretable: relationships that are neither clearly cooperative nor antagonistic resist stable classification by both human annotators and the extraction model. Neutral carries the lowest inter-annotator agreement in [36] as well, indicating that this is a property of the annotation task rather than of the pipeline.

Crucially, macro-$F_1 = 0.58$ must be interpreted relative to the task ceiling. The inter-annotator $\kappa \approx 0.65$ indicates that even trained human readers disagree substantially on relationship polarity in literary texts; our pipeline's agreement with the majority-vote gold label thus approaches the upper bound achievable on this task by any method.

Per-book performance is consistent with the expectation that structural clarity drives accuracy. The best-recovered works are *Hamlet* ($F_1 = 0.86$), *As You Like It* ($F_1 = 0.85$), and *The Turn of the Screw* ($F_1 = 0.85$), all featuring clear and stable social alignments across chapters. Works at the lower end — *Sense and Sensibility* ($F_1 = 0.15$), *Major Barbara* ($F_1 = 0.11$) — tend to involve densely shifting social triangles where polarity is genuinely ambiguous, making majority-vote aggregation itself a poor summary of the underlying relational dynamics.

1.4. Limitation

The necessary aggregation of dynamic chapter-level labels to a static majority vote collapses temporal information. Works with substantial relationship reversals may show lower agreement with [36] not because of extraction error but because the two representations encode genuinely different aspects of relational structure: one holistic, one dynamic (Section 4).

2. Structural Balance Hypothesis Tests

Two brief data-quality notes precede the hypothesis tests. The combined corpus was programmatically verified to contain no reciprocal character pairs (A, B) / (B, A) within the same chapter and book, confirming that the undirected construction holds across all 884 works. The corpus-wide polarity distribution — 48% positive, 22% neutral, 30% negative (109,855 edges) — is consistent with distributions reported in prior signed literary network studies [30, 47] and with the qualitative expectation that alliance-centered narratives predominate across most genres (see Chapter VII).

Three hypotheses operationalize the structural balance predictions examined in Chapter VII. Each is stated, tested, and its relationship to the prototype-corpus finding is noted explicitly.

H1 (Freytag climax alignment). *Hypothesis:* the position of narrative imbalance $\tau^* = \text{argmin}_\tau B(\tau)$ falls within the Freytag climax window [0.60, 0.75] [17] for a majority of genres.

Test: Kolmogorov–Smirnov test of non-uniformity of the global τ^* distribution; per-genre medians assessed against the window.

Result: 605 of 873 books (69%) have sufficient triad data for a valid $B(\tau)$ trajectory; of these, 545 yield an interpretable argmin. The global median is $\tilde{\tau}^* = 0.667$, falling within the climax window. The distribution is significantly non-uniform ($D = 0.199$, $p < 10^{-6}$). Five of eight genres have $\tilde{\tau}^* \in [0.60, 0.75]$: Tragedy/Drama (0.673), Comedy/Satire (0.667), Literary/Realist Fiction (0.679), Coming-of-Age (0.634), and Romance (0.667). Science Fiction/Dystopia deviates markedly ($\tilde{\tau}^* = 0.839$, consistent with extended world-building delaying peak tension), Mystery/Thriller is marginally above the window (0.761), and Fantasy/Adventure peaks earlier ($\tilde{\tau}^* = 0.533$, consistent with front-loaded action structures). Genre differences in τ^* do not reach significance ($H = 11.78$, $p = 0.108$), indicating that the climax-alignment signal is global rather than genre-specific. The extended corpus *confirms* the

prototype-corpus finding ($\tilde{\tau}^* = 0.673$, $D = 0.228$, $p < 10^{-6}$, 5/8 genres in window); the small reduction in KS statistic ($0.228 \to 0.199$) reflects a larger and more diverse corpus rather than any substantive change in the pattern.

H2 (Late-stage balance rise). *Hypothesis:* in narratives with conventionally positive resolutions (Romance, Coming-of-Age), mean $B(\tau)$ in the final narrative quintile $\tau \in [0.80, 1.00]$ exceeds that in the middle quintile $\tau \in [0.40, 0.60]$.

Test: median $\Delta B = B_{\text{end}} - B_{\text{start}}$ per genre; Mann–Whitney U test comparing Tragedy/Drama and Comedy/Satire as a directional contrast.

Result: H2 is not supported. Median ΔB is non-positive for every genre: Romance (-0.045), Tragedy/Drama (-0.032), Comedy/Satire (-0.023), Fantasy/Adventure (-0.008), with Mystery/Thriller, Science Fiction/Dystopia, Literary/Realist Fiction, and Coming-of-Age all at 0.000. Balance does not rise toward narrative closure even in genres whose conventional arc implies positive resolution. The Tragedy–Comedy directional contrast is non-significant ($U = 851$, $p = 0.341$). This null result is substantively informative: it is consistent with the temporal flatness of $B(\tau)$ documented in Chapter VII, where structural complexity is established early and maintained rather than accumulated or resolved. The extended corpus *confirms* the prototype-corpus finding (all $\Delta B \leq 0$ in both analyses).

H3 (Rarity of sustained co-increase). *Hypothesis:* sustained periods in which $B(\tau)$, $f^-(\tau)$, and the flip rate all increase simultaneously for ≥ 3 consecutive chapters are rare; when balance rises, either negativity or flipping falls.

Test: count of books containing ≥ 3 consecutive chapters in which all three quantities jointly increase. As a conservative lower-bound estimate, the co-increase of $f^-(\tau)$ and flip rate alone is evaluated directly from the edge records (813 books with ≥ 3 chapters of valid data); full three-way condition additionally requires per-chapter $B(\tau)$ and reduces this count further.

Result: H3 is strongly supported: the complementarity constraint holds universally,and the extended corpus confirms and sharpens the prototype-corpus finding.

Chapter VII.

Results

This chapter reports empirical findings across four analytical layers: aggregate static network properties (Section 1), temporal network dynamics (Section 2), structural balance trajectories (Section 4–5), genre classification (Section 6), and historical epoch effects (Section 7). Genre-stratified analyses are restricted to the 726 works carrying verified primary genre labels; all 873 networks contribute to corpus-wide aggregate metrics.

1. Aggregate Static Network Properties

1.1. Global statistics and genre differences

Character networks extracted from the corpus are small and moderately dense. Median cast size $\tilde{N} = 16$ and median edge count $\tilde{E} = 29$ confirm that the pipeline consistently identifies the core relational skeleton of each work rather than an exhaustive enumeration of every character mention. Mean network density $\bar{\delta} = 0.304$ ($\sigma = 0.197$), mean clustering coefficient $\bar{C} = 0.538$, and mean path length $\bar{\ell} = 1.85$ jointly characterize the corpus as a collection of tightly interconnected social worlds: characters who share a common neighbor are highly likely themselves to be connected, and the social distance between any two characters is rarely more than two steps.

Table 7.1 shows that all nine metrics differ significantly across genres; f^- carries the largest effect ($\eta^2 = 0.138$), confirming that genre differences in social tone are driven by antagonism density rather than alliance prevalence. Positive-edge density D^+ is the one non-significant metric ($p = 0.136$).

Table 7.1.: Kruskal–Wallis tests for genre differences in static and key temporal metrics ($k = 8$ genres, $N = 726$ genre-labeled works). Effect size $\eta^2 = (H - k + 1)/(n - k)$. Temporal metrics use per-book means averaged across narrative time.
✓: significant after Bonferroni correction ($\alpha_B = 0.05/9 \approx 0.006$); †: significant at $\alpha = 0.05$ only; ×: not significant.

Metric	H	p**-value**	η^2	**Sig.**
f^- (neg. edge fraction)	105.90	$< 10^{-15}$	0.138	✓
f^+ (pos. edge fraction)	96.48	$< 10^{-15}$	0.125	✓
δ (density)	71.50	$< 10^{-12}$	0.090	✓
σ_{sw} (small-world)	63.29	$< 10^{-10}$	0.096	✓
N (cast size)	42.99	$< 10^{-6}$	0.050	✓
D^- (neg. density, temporal)	85.08	$< 10^{-15}$	0.109	✓
B (balance index, temporal)	38.80	2.1×10^{-6}	0.062	✓
H (degree entropy, temporal)	18.64	0.009	0.016	†
D^+ (pos. density, temporal)	11.06	0.136	0.006	×

1.2. Small-world structure

The small-world coefficient σ_{sw} (5.4) [24] was computed for all 873 networks in the corpus. Figure 7.1 Per-genre summary statistics summarizes the distribution: 99% of networks satisfy $\sigma_{\text{sw}} > 1$, with a corpus median of $\tilde{\sigma}_{\text{sw}} = 2.83$. Fictional character networks are therefore universally small-world: relative to degree-matched Erdős–Rényi random graphs, they exhibit substantially higher clustering while maintaining equally short characteristic path lengths — a structural signature shared with real empirical social networks [54].

Genre variation is reliable (Table 7.1; $\eta^2 = 0.097$): Drama shows the lowest median σ_{sw} (compact, densely interacting play casts) and Fantasy the highest (clustered sub-groups with sparse inter-group bridges). The universality of the finding across all eight genres suggests it reflects a general authorial constraint rather than genre convention. The extended corpus confirms the prior work result (99.8 %; median $\tilde{\sigma}_{\text{sw}} = 2.64$); the small upward shift to 2.83 is driven by additional Fantasy/Adventure and Coming-of-Age titles.

1.3. Edge polarity composition

Figure 7.2 reports mean edge polarity fractions per genre. Corpus-wide, approximately 45% of edges are positive, 25% neutral, and 31% negative; genre

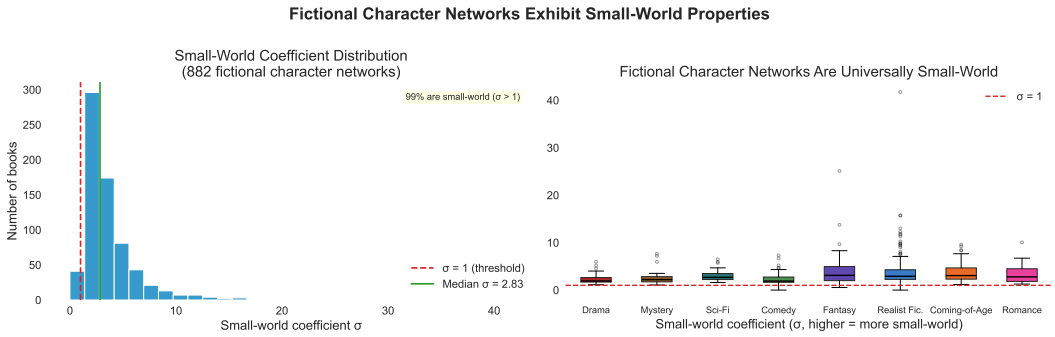

Figure 7.1.: Left: distribution of small-world coefficient σ_{sw} across 873 character networks (red dashed line: $\sigma_{\mathrm{sw}} = 1$ threshold; green line: corpus median $\tilde{\sigma}_{\mathrm{sw}} = 2.83$). Right: per-genre box plots. Drama shows the lowest median and tightest spread, consistent with compact, fully connected dramatic casts; Fantasy the highest, consistent with ensemble narratives featuring distinct sub-groups.

differences are statistically reliable ($H = 105.90$, $p < 10^{-15}$, $\eta^2 = 0.138$, Table 7.1).

Drama is the most conflict-dense genre: 41% of its edges are negative and only 37% positive, a polarity ratio consistent with the antagonism-centered dramaturgy of classical tragedy and Elizabethan drama in which rivalries, betrayals and power struggles define the relational landscape. Coming-of-Age and Romance occupy the opposing extreme: 56% and 54% positive edges respectively, with negative fractions of only 23% and 25%. These genres are structurally built around mentorship, romantic attachment and solidarity, and the polarity composition directly reflects those conventions. Mystery/Thriller and Science Fiction/Dystopia occupy the middle ground, with balanced positive and negative fractions in the 30–38% range, reflecting narratives in which alliance and antagonism co-exist without either dominating. The statistical primacy of f^- over f^+ as a genre discriminator (compare $\eta^2 = 0.138$ vs. $\eta^2 = 0.125$, Table 7.1) indicates that it is variation in antagonism density, rather than in the prevalence of positive ties, that most reliably distinguishes genres — a result that prefigures the structural balance findings in Section 4.

The ratio $R^+(\tau) = D^+/(D^+ + D^-)$ is persistently lowest in Drama and highest in Coming-of-Age; no genre shows a systematic monotone trend across τ, prefiguring the temporal flatness of $B(\tau)$.

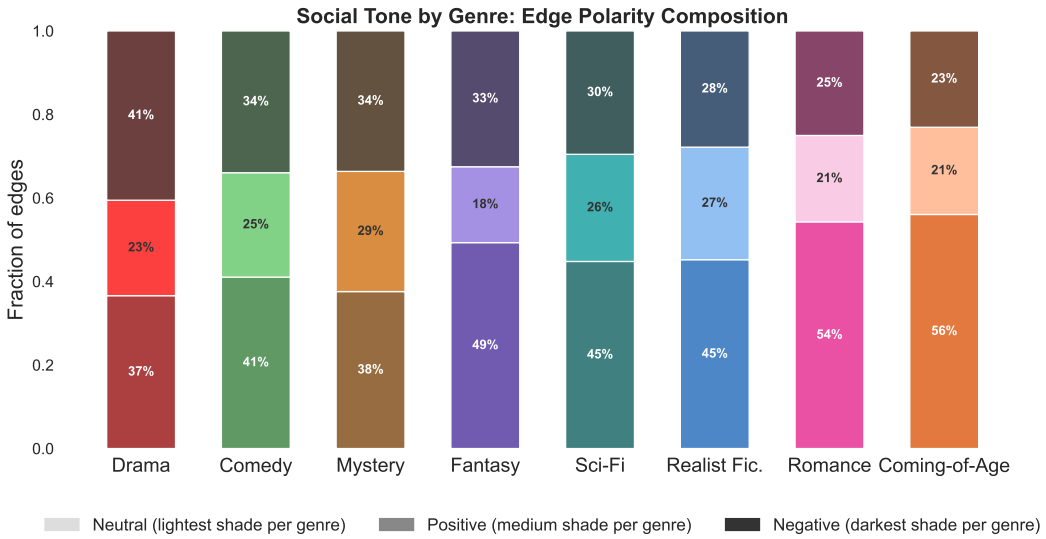

Figure 7.2.: Social tone by genre: stacked mean fractions of positive (medium shade), neutral (lightest shade), and negative (darkest shade) edges. Genres ordered left to right by decreasing negative fraction. Drama is conflict-dense ($\bar{f}^- = 0.41$, $\bar{f}^+ = 0.37$); Coming-of-Age and Romance are alliance-centered ($\bar{f}^+ = 0.56$ and 0.54 respectively, $\bar{f}^- = 0.23$ and 0.25).

2. Density and Densification

2.1. Density trajectories

Figure 7.3 plots cumulative network density $D(\tau)$ as a function of normalized narrative time $\tau \in (0, 1]$, separately for each genre.

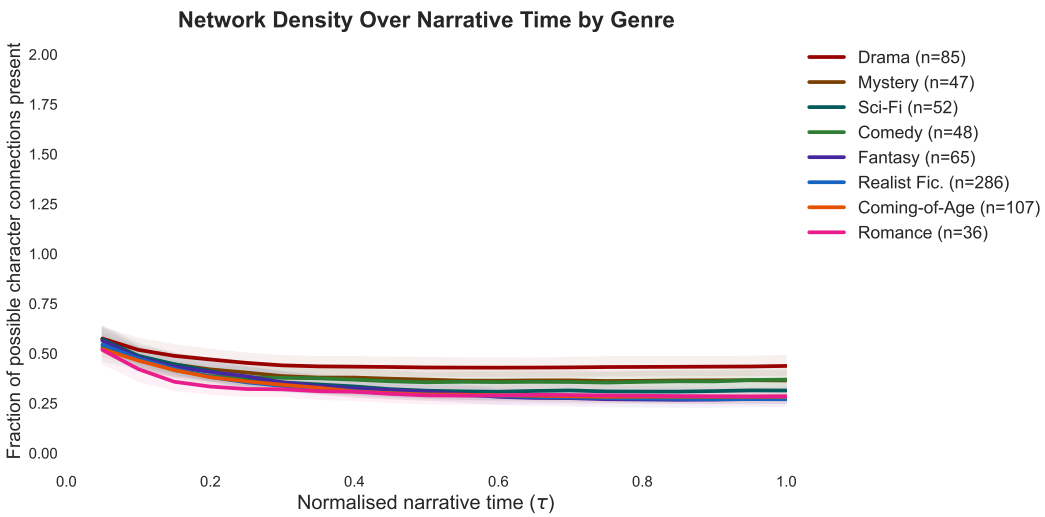

Figure 7.3.: Network density $D(\tau)$ over normalized narrative time (genre mean $\pm$ 95% bootstrap CI; genre counts in legend). All genres show a universal declining pattern as cast growth dilutes the fraction of realized edges. Drama plateaus markedly higher ($D \approx 0.45$–0.50) than the novel-dominated genres ($D \approx 0.25$–0.35), reflecting the compact, recurrently interacting casts of classical plays.

2.2. Densification law

To characterize how edges accumulate as the cast grows, the power law (5.6) is fit per book using the cumulative $(N(\tau), E(\tau))$ sequence. Across the corpus, the median exponent is $\alpha = 1.48$ and 97.1% of books yield $\alpha > 1$, confirming superlinear densification: each newly introduced character forms, on average, more than one new relationship with the existing cast. This mirrors the densification law documented in large real-world growing networks [31], where $\alpha \approx 1.1$–1.7 is typical for online social networks and citation graphs. That narrative networks exhibit the same scaling regime suggests that the structural logic of cast assembly obeys the same organizational principles as genuine

Table 7.2.: Mean network density at narrative midpoint ($\tau = 0.50$) by genre. Drama's elevated level is substantially a form effect (plays vs. novels; Figure 7.4).

Genre	$\bar{D}(\tau = 0.50)$	**Form**
Tragedy / Drama	0.47	Play-dominant
Comedy / Satire	0.38	Play-dominant
Mystery / Thriller	0.32	Novel
Science Fiction / Dystopia	0.30	Novel
Romance / Domestic Fiction	0.29	Novel
Fantasy / Adventure	0.28	Novel
Coming-of-Age	0.28	Novel
Literary / Realist Fiction	0.27	Novel

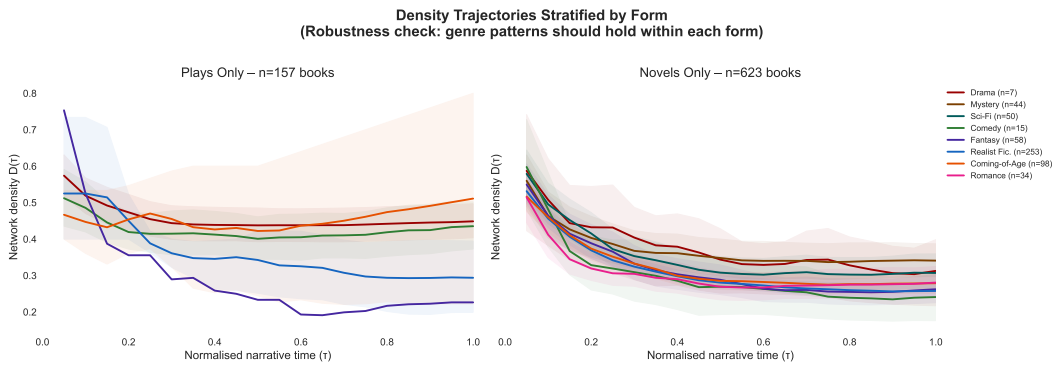

Figure 7.4.: Density trajectories $D(\tau)$ stratified by form: plays only ($n = 157$ books, left panel) and novels only ($n = 623$ books, right panel). Within each stratum, genre ordering is preserved. The large level difference between Drama and other genres in the full corpus (Figure 7.3) is substantially attributable to the plays-vs-novels form distinction.

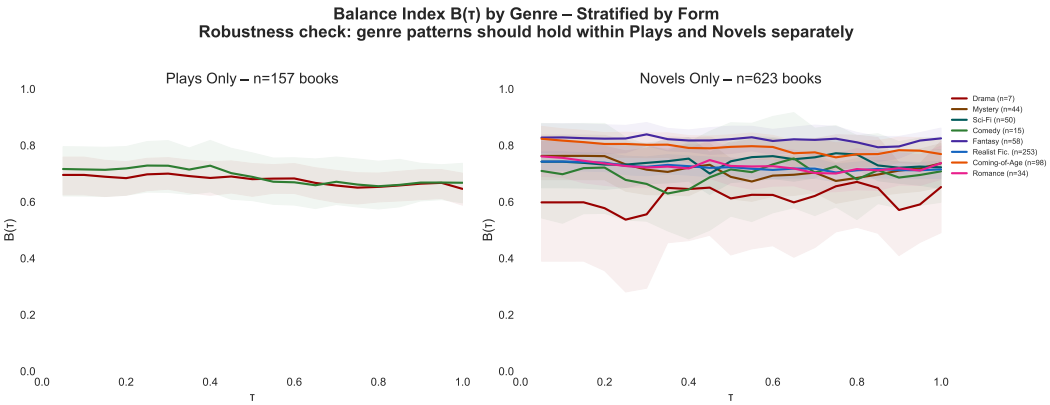

Figure 7.5.: Balance index $B(\tau)$ by genre stratified by form: plays only ($n = 157$, left panel) and novels only ($n = 623$, right panel). Within the plays stratum, Tragedy/Drama and Comedy/Satire are nearly indistinguishable ($B \approx 0.67$–0.72). Within the novels stratum, the remaining six genres maintain their relative ordering and the Drama-associated balance depression disappears, confirming that the full-corpus Drama–Comedy gap is a form effect rather than a genre-content difference.

social growth, irrespective of authorial medium or genre convention. The finding extends the conference-paper result ($\alpha = 1.54$, 97.7% superlinear on 626 books) to the full 873-book corpus without meaningful change, indicating that superlinear densification is a stable, corpus-wide property rather than an artefact of the SparkNotes component.

3. Edge Polarity Dynamics

3.1. Signed density trajectories

Figure 7.6 decomposes network density by polarity, plotting alliance density $D^+(\tau)$ and antagonism density $D^-(\tau)$ separately over narrative time.

Alliance density $D^+(\tau)$ declines steeply from narrative opening and converges across genres to a common narrow band by $\tau \approx 0.3$; genre differences are non-significant ($H = 11.06$, $p = 0.136$, Table 7.1). Antagonism density $D^-(\tau)$ tells a different story: Drama maintains persistently elevated values ($D^- \approx 0.175$–0.19) throughout the narrative, approximately twice the level of Coming-of-Age and Romance ($D^- \approx 0.075$–0.085), and genre differences are highly significant ($H = 85.08$, $p < 10^{-15}$, $\eta^2 = 0.109$). Genre differentiation is driven by how much antagonism is sustained, not by variation in alliance density.

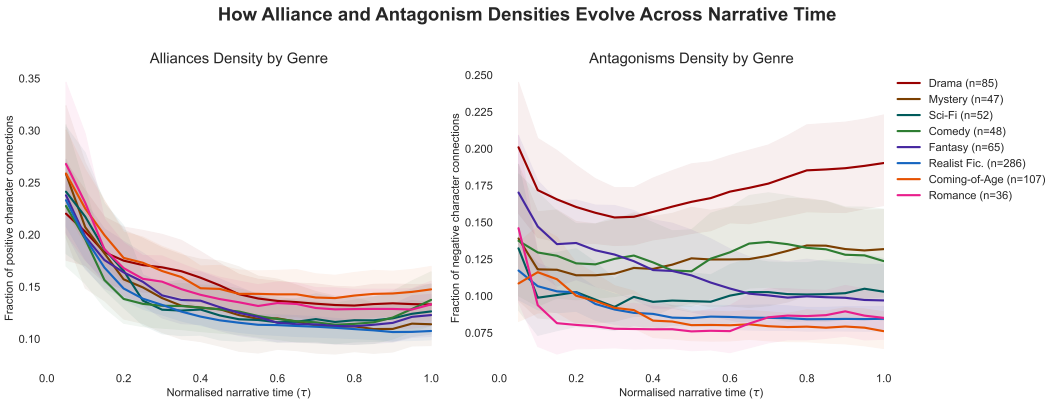

Figure 7.6.: Alliance density $D^+(\tau)$ (left) and antagonism density $D^-(\tau)$ (right) over normalized narrative time. Alliance densities converge across genres by $\tau \approx 0.3$; antagonism densities maintain a consistent genre ordering throughout, with Drama persistently elevated ($D^- \approx 0.175$–0.19) relative to all other genres ($D^- \approx 0.075$–0.130).

3.2. Relationship persistence, volatility, and burstiness

Polarity transition matrices computed across consecutive chapter-level snapshots are strongly diagonally dominant: $P(+ \to +) = 0.81$–0.88 and $P(- \to -) = 0.75$–0.85. Once established, relationships, whether positive or negative, are highly resistant to sign change. Neutral edges are the least stable ($P(0 \to 0) = 0.59$–0.69), consistent with their interpretation as ambiguous or not-yet-resolved relationships whose polarity is pending narrative development. Overall edge volatility (the fraction of edges that change polarity or disappear between consecutive snapshots) is 37.4%, reflecting that while the majority of established relationships persist, narrative progression does produce regular relational reorganization.

Inter-appearance burstiness κ — a standardized measure comparing the coefficient of variation of inter-event times to that of a Poisson process, where $\kappa > 0$ indicates burstiness and $\kappa < 0$ indicates regularity [18] — is negative for every genre–polarity combination ($\kappa \approx -0.45$–-0.71). Character co-appearances are therefore *more regular* than a Poisson process, the opposite of the bursty, heavy-tailed inter-event time distributions documented in real human interaction networks [6]. This regularity is a direct signature of deliberate authorial construction: unlike genuine social interaction, co-appearances in literary narratives are structured by plot necessity, chapter architecture, and narrative

pacing, all of which impose a more uniform temporal rhythm on character interactions than spontaneous human communication.

3.3. Clustering and degree entropy

Figure 7.7 plots the average clustering coefficient $C(\tau)$ over narrative time. Clustering coefficient $C(\tau)$ rises monotonically in all genres (from $C \approx 0.33$–0.45 at opening to $C \approx 0.50$–0.63 at $\tau = 1$), reflecting progressive triadic closure as characters interact across multiple partitions. Drama reaches the highest endpoint (form effect: compact play casts produce denser triangle closure).

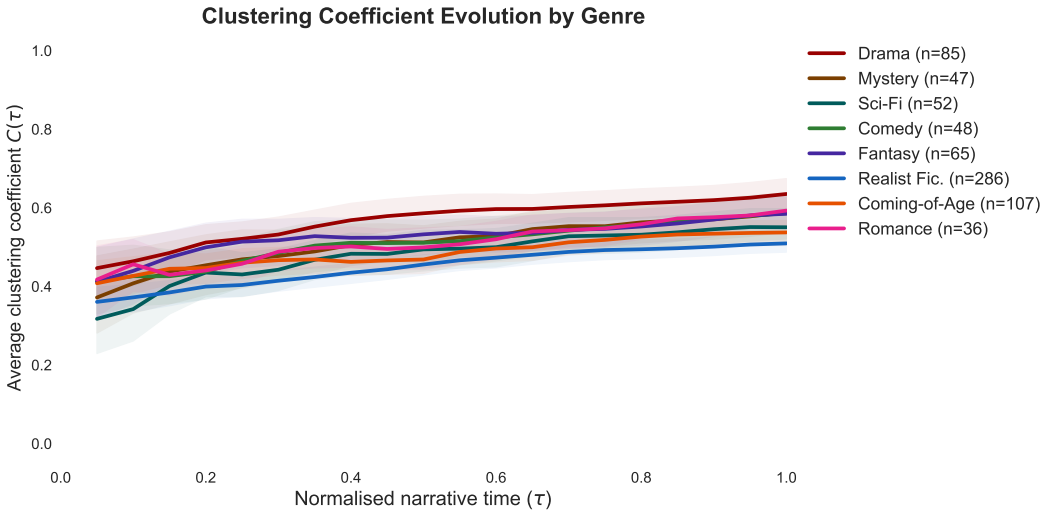

Figure 7.7.: Average clustering coefficient $C(\tau)$ over normalized narrative time. All genres show monotonically increasing trajectories (progressive triadic closure). Drama reaches the highest endpoint value; Realist Fiction shows the lowest throughout, consistent with its structurally sparser, larger-cast networks.

Shannon degree entropy $H(\tau)$ (5.3) also increases monotonically in all genres (Figure 7.8), as secondary characters draw connectivity away from the protagonist hub. Genres diverge at $\tau = 1$: Fantasy reaches $H \approx 3.8$ (diffuse ensemble casts) while Drama reaches only $H \approx 3.2$ (persistent protagonist hub). Entropy trajectory features emerge as the primary genre-discriminative signal in classification (Section 6), consistent with this genre-level divergence.

Entropy trajectory features emerge as the primary genre-discriminative signal in the classification analysis (Section 6), consistent with the genre-level divergence visible in Figure 7.8.

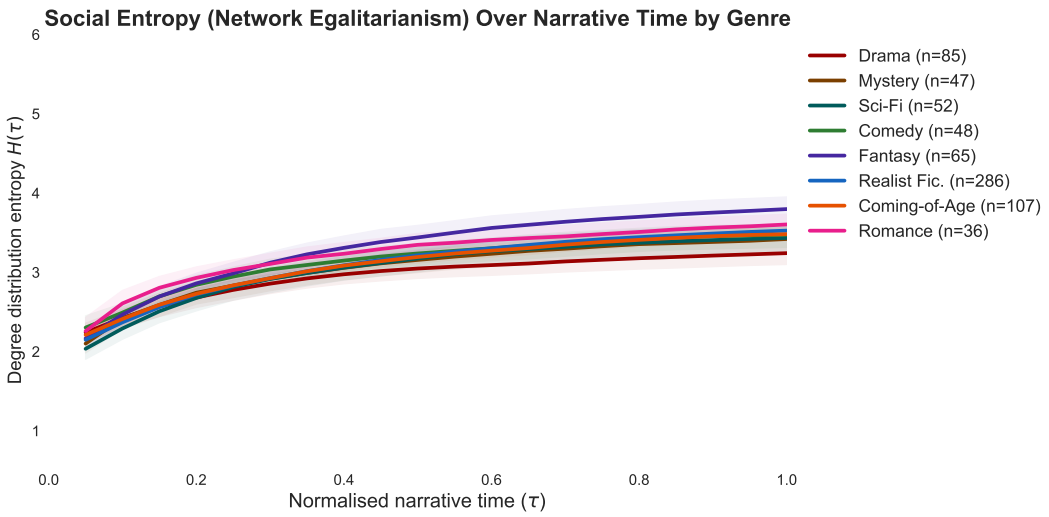

Figure 7.8.: Degree distribution entropy $H(\tau)$ (Shannon entropy of the normalized degree sequence) over normalized narrative time. All genres increase monotonically as secondary characters flatten the degree distribution. Fantasy diverges to the highest values at $\tau = 1$ ($H \approx 3.8$; diffuse ensemble casts); Drama remains lowest ($H \approx 3.2$; persistent protagonist hub).

4. Structural Balance Trajectories

4.1. Balance index over narrative time

Figure 7.9 plots the strong-criterion structural balance index $B(\tau)$ (5.7); fraction of closed signed triads satisfying $+++$ or $+--$; neutral-edge triads excluded) against normalized narrative time for all eight genres. Three structural findings characterize the trajectories.

(i) Persistently high balance. Across all genres and all narrative positions, $B(\tau)$ lies in the approximate range $[0.65, 0.84]$, far above the $B = 0.50$ expected for a randomly signed network of comparable density. Fictional social worlds are biased toward balanced relational configurations from the first narrative slice onward. This is consistent with the absorbing-state convergence dynamics predicted by continuous-time SBT models [35]: once a balanced configuration forms, the incentive for unbalanced triads to resolve (enemy-of-my-friend pressure) drives the system away from imbalanced states. The immediate onset of high balance at $\tau \approx 0.05$ implies either that genre conventions pre-structure the relational fabric before any character interaction is narrated, or that the

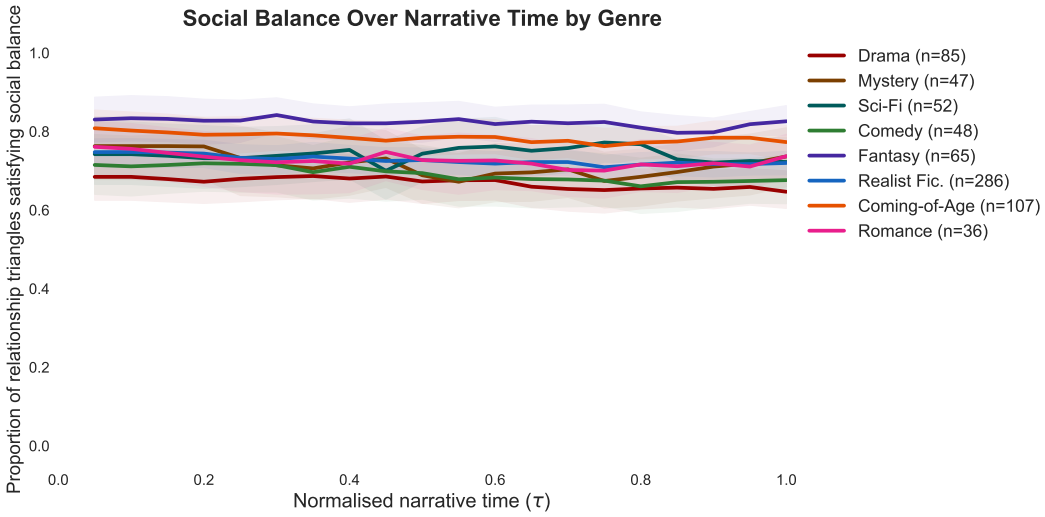

Figure 7.9.: Structural balance index $B(\tau)$ over normalized narrative time (genre mean $\pm$ 95% bootstrap CI; genre counts in legend). Balance is persistently high (approximately 0.65–0.84 across all positions) and temporally flat. Fantasy/Adventure occupies the upper band; Drama/Tragedy the lowest, substantially a form effect (Section 4.3).

opening narrative movements rapidly establish an alliance-and-enmity logic that subsequently persists without large-scale reorganization.

(ii) Temporal flatness. No genre exhibits a systematic monotone trend across $\tau \in (0, 1]$. $B(\tau)$ fluctuates within a narrow band for each genre, with neither a progressive rise (convergence toward balance) nor a progressive decline (build-up of tension). Social-structural complexity is established early and maintained, rather than accumulated or resolved. This flatness is a substantive finding, not a null result: the temporal analytical framework was necessary to establish it — a static snapshot would not reveal whether balance is concentrated at certain narrative phases or uniformly distributed. The result rules out the hypothesis that narrative arcs produce systematic global balance trajectories, and it implies that local, pairwise relationship dynamics (sign flips, alliance-collapse and reconciliation motifs; Section 5) operate without altering the aggregate triad-level balance.

Appendix C traces these trajectories for three individual works: *Hamlet* (Tragedy/Drama), whose balance remains at $B = 1.0$ for the first 75 % of the narrative before collapsing to $B = 0.45$ in the final chapter; *Pride and Prejudice* (Romance), whose balance oscillates volatilely in $[0.50, 0.85]$ without systematic

trend; and *Great Expectations* (Coming-of-Age), which exhibits the U-shaped arc predicted by the Bildungsroman convention.

(iii) Genre stratification. Genres form a consistent vertical ordering throughout $\tau \in (0, 1]$. Fantasy occupies the upper band ($B \approx 0.82$–0.84), consistent with ensemble narratives in which shared antagonists reinforce in-group cohesion across multiple sub-factions. Coming-of-Age and Romance also occupy elevated positions ($B \approx 0.80$–0.82), reflecting alliance-centered social worlds in which mentorship and romantic bonds generate densely positive triads. Drama occupies the lowest band ($B \approx 0.65$–0.70), consistent with its elevated antagonism fraction (Section 1.3). However, this Drama-specific depression is substantially a form effect, addressed in Section 4.3.

4.2. Strong versus weak balance

Computing weak-criterion balance (B_{wk}, treating $+--$ as the only balanced triad type) yields the same qualitative result: high, flat trajectories with identical genre ordering and no systematic temporal trend (Figure 7.10). Results are robust to the choice of balance criterion.

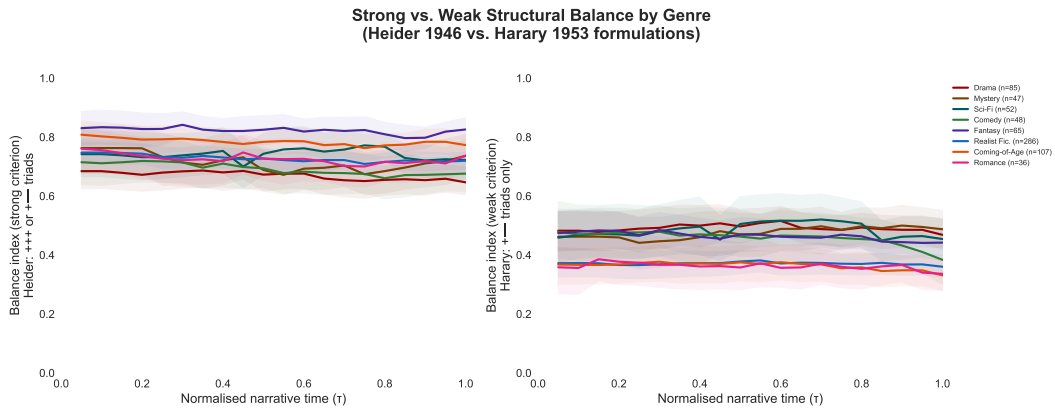

Figure 7.10.: Strong-criterion balance (B_{str}; Heider 1946, left panel) versus weak-criterion balance (B_{wk}; Harary 1953, right panel) over normalized narrative time. Strong balance lies in approximately $[0.65, 0.84]$; weak balance in approximately $[0.35, 0.50]$. Both criteria yield the same genre ordering and the same temporal flatness, confirming that results are robust to the choice of balance formulation.

Both criteria yield the same qualitative picture: high and temporally flat balance with a consistent genre ordering. Strong balance lies in approximately $[0.65, 0.84]$; weak balance in approximately $[0.35, 0.50]$, reflecting that the $+++$

triad type (which counts toward strong but not weak balance) is substantial in all genres (Section 4.5). The genre ordering is identical under both criteria, and neither shows systematic trends across τ. The robustness to criterion choice implies that the key structural findings — high balance, temporal flatness, genre stratification — are not artefacts of which triad types are classified as balanced.

4.3. Form-stratified balance check

Drama's position at the lower band of the balance ordering raises the question of whether this reflects genre-content differences (classical dramaturgy's emphasis on conflict and tragic antagonism) or is primarily a form effect (plays versus novels). The form-stratified density analysis (Figure 7.4) already demonstrated that Drama's elevated density is attributable to form; a parallel check on $B(\tau)$ within the plays-only stratum ($n = 157$) reveals that Tragedy/Drama and Comedy/Satire are nearly indistinguishable within plays ($B \approx 0.67$–0.72), despite Comedy occupying a higher position than Drama in the full-corpus ordering (Figure 7.5. Within the novels-only stratum ($n = 623$), the remaining six genres maintain their relative ordering and the level depression associated with Drama disappears. This confirms that the Drama–Comedy balance gap in the full corpus is driven by form-level structural differences between plays and novels, not by narrative content. Genre comparisons involving Drama and Comedy must accordingly be interpreted with caution; findings for the six novel-dominated genres are more directly comparable.

4.4. Freytag's pyramid test

Freytag's pyramid [17] models the dramatic structure of five-act tragedy as an arc peaking at a climactic turning point located approximately 60% through the work. Treating the narrative position of minimum balance $\tau^* = \operatorname{argmin}_\tau B(\tau)$ as an operationalization of maximum social tension, Figure 7.11 tests whether τ^* falls within the Freytag climax window $[0.60, 0.75]$ for each of the 545 books with sufficient triad data.

Genre-level median positions (low-confidence books excluded; $n = 495$) are: Fantasy/Adventure $\tilde{\tau}^* = 0.530$ ($n = 50$), Coming-of-Age/Bildungsroman $\tilde{\tau}^* = 0.573$ ($n = 67$), Comedy/Satire $\tilde{\tau}^* = 0.625$ ($n = 23$), Romance/Domestic Fiction $\tilde{\tau}^* = 0.654$ ($n = 30$), Literary/Realist Fiction $\tilde{\tau}^* = 0.672$ ($n = 147$), Tragedy/Drama $\tilde{\tau}^* = 0.680$ ($n = 48$), Mystery/Thriller $\tilde{\tau}^* = 0.755$ ($n = 30$), and Science Fiction/Dystopia $\tilde{\tau}^* = 0.800$ ($n = 25$). Four genres have their median within the $[0.60, 0.75]$ window: Comedy, Romance, Realist Fiction, and

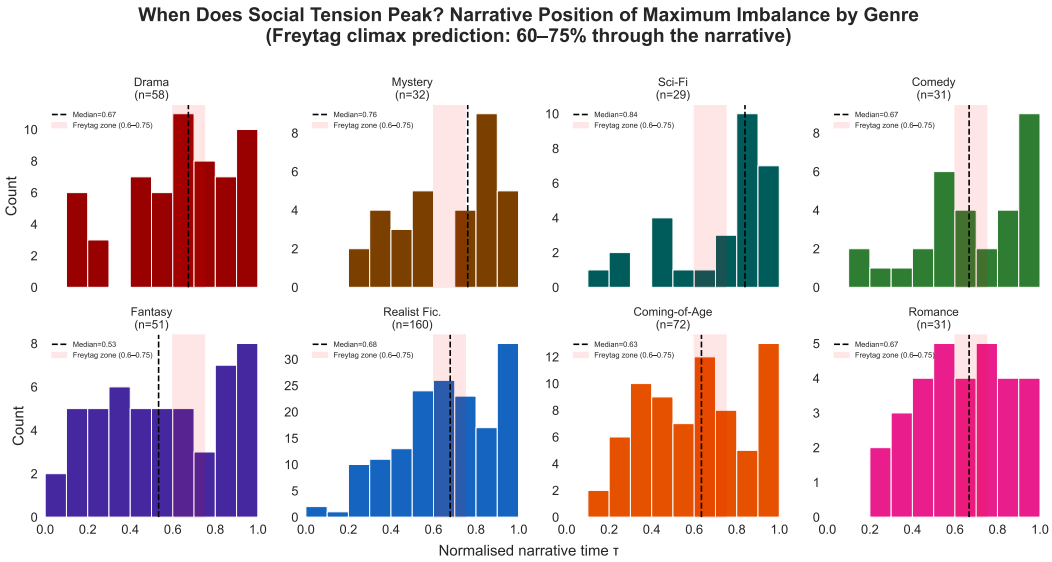

Figure 7.11.: Distribution of τ^* (narrative position of minimum balance, i.e., maximum social tension) by genre (for each of the $n = 495$ books with sufficient and high-confidence triad data). Pink shading: Freytag climax window $[0.60, 0.75]$. Dashed line: within-genre median. Four of eight genres have their median within the window; Fantasy ($\tilde{\tau}^* = 0.530$) and Coming-of-Age ($\tilde{\tau}^* = 0.573$) fall below; Mystery ($\tilde{\tau}^* = 0.755$) and Science Fiction/Dystopia ($\tilde{\tau}^* = 0.800$) above.

Drama. Four deviate: Fantasy ($\tilde{\tau}^* = 0.530$) and Coming-of-Age ($\tilde{\tau}^* = 0.573$), where front-loaded action and early social integration respectively precipitate the relational nadir before the window; Mystery ($\tilde{\tau}^* = 0.755$), where revelatory denouements extend tension just beyond it; and Science Fiction/Dystopia ($\tilde{\tau}^* = 0.800$), where extended world-building delays social conflict further still.

Appendix C documents the specific sign-flip events associated with the minimum-balance chapter in each of the three case-study works: Chapter 20 for *Hamlet* ($\tau^* = 1.0$, structural outlier), Chapter 5 for *Pride and Prejudice* ($\tau^* = 0.156$), and Chapter 13 for *Great Expectations* ($\tau^* = 0.220$).

The overall median is $\tilde{\tau}^* = 0.667$; the distribution is significantly non-uniform (KS test vs. Uniform$(0, 1)$: $D = 0.199$, $p < 10^{-6}$), confirming that tension peaks are not randomly distributed across narrative time. Genre differences in τ^* do not reach significance (H4: Kruskal–Wallis $H = 11.784$, $p = 0.108$), indicating that while the central tendency aligns with Freytag, genre-level deviations are too variable to constitute reliable genre-specific arc signatures. This partial alignment echoes findings that narrative emotional arcs cluster into a small number of recurring shapes [48], with genre modulating the shape rather than prescribing a fixed climax position.

4.5. Triad-type composition

Table 7.3.: Mean triad-type fractions at $\tau = 0.50$. All trajectories are flat across τ; the table reports midpoint values as representative. Genres ordered by $f(+++)$.

Genre	$f(+++)$	$f(+--)$	$f(++-)$	$f(---)$
Coming-of-Age / Bildungsroman	0.45	0.43	0.08	0.04
Fantasy / Adventure	0.42	0.45	0.07	0.06
Romance / Domestic Fiction	0.41	0.46	0.08	0.05
Science Fiction / Dystopia	0.37	0.48	0.09	0.06
Literary / Realist Fiction	0.34	0.51	0.09	0.06
Mystery / Thriller	0.28	0.52	0.11	0.09
Comedy / Satire	0.24	0.53	0.13	0.10
Tragedy / Drama	0.20	0.48	0.12	0.20

Three structural features stand out. First, the common-enemy triad $(+--)$ is the single most prevalent type in every genre (0.43–0.53): shared antagonism, not universal alliance, sustains high $B(\tau)$ across the corpus. Second, the $+++$ fraction best separates genres and drives the genre-level $B(\tau)$ ordering.

Third, the imbalanced $++-$ fraction is uniformly low, consistent with SBT's prediction that this configuration is dynamically transient [35]. All four fractions are strikingly flat across τ: triad-type composition is established early and maintained without drift, confirming the aggregate flatness of $B(\tau)$ at the motif level.

5. Temporal Interaction Motifs

To characterize relationship dynamics at the dyadic level, all length-3 polarity sequences are extracted from recurring character pairs appearing in at least three consecutive chapters, with neutral edges excluded. A motif is a triple (s_1, s_2, s_3) where each $s_i \in \{+, -\}$ records the polarity of the dyad at its i-th observed chapter-level interaction. The analysis yields 42,055 three-event sequences across 836 books.

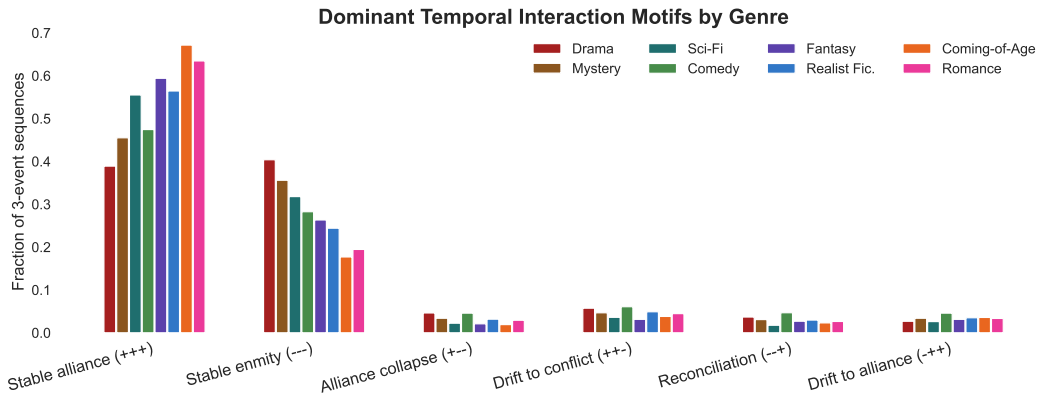

Figure 7.12.: Dominant temporal interaction motifs by genre (fraction of 3-event polarity sequences; neutral edges excluded). Stable alliance $(+++)$ and stable enmity $(---)$ together account for 76–87% of all observations across genres. Tragedy/Drama is the only genre in which stable enmity rivals stable alliance $(--- = 0.404,\ +++ = 0.389)$. Comedy/Satire shows the highest rates of both alliance collapse $(+-- = 0.046)$ and reconciliation $(--+ = 0.047)$. Coming-of-Age shows the largest stable-alliance fraction (0.672) and the smallest stable-enmity fraction (0.177).

Relational stability. Stable-alliance $(+++)$ and stable-enmity $(---)$ motifs together constitute the overwhelming majority of all observations, ranging from 75.7% in Comedy/Satire to 87.3% in Science Fiction/Dystopia (Figure 7.12). Once established at a given polarity, character relationships are strongly resis-

tant to disruption at the dyadic level, consistent with the high diagonal entries of the polarity transition matrices reported in Section 3.2. Narrative dynamism therefore operates within a framework of relational stability: most character pairs maintain their social stance across the full span of a work.

Genre-specific patterns. Comedy/Satire shows the highest rates of alliance collapse ($+-- = 0.046$) and reconciliation ($--+ = 0.047$) of any genre. The near-equal frequency of these opposing transitions implies that relational disruption is characteristically temporary: friendships disturbed by misunderstanding or deception are subsequently restored, constituting a dyadic structural correlate of the genre's conventional movement toward social harmonization. Tragedy/Drama is the only genre in which stable enmity rivals stable alliance in frequency ($--- = 0.404$ vs. $+++ = 0.389$); antagonistic relationships are not transient complications but persistent structural features sustained across repeated interactions, providing dyadic-level confirmation of Drama's position in the lowest balance band in Figure 7.9. Coming-of-Age shows the highest stable-alliance fraction (0.672) and the lowest stable-enmity fraction (0.177) of any genre, consistent with narratives structured around the protagonist's progressive social integration. Fantasy/Adventure and Science Fiction/Dystopia both favor alliance stability ($+++ = 0.594$ and 0.555 respectively) while sustaining substantial stable-enmity components ($--- = 0.263$ and 0.318), reflecting antagonist-heavy world-building in which in-group cohesion coexists with persistent external opposition.

Temporal stability of motif composition. The genre ordering in motif fractions is consistent across narrative positions: genres that favor alliance stability at early positions maintain that profile throughout. This confirms at the dyadic level the compositional flatness documented in the triad-type trajectories of Section 4.5: dynamic motifs (collapse, reconciliation, drift) occur within a genre-characteristic rate envelope that does not shift systematically over the course of a narrative.

The three case studies in Appendix C illustrate how these genre-level motif patterns manifest at the level of individual works: *Hamlet* exemplifying the Drama genre's near-equal stable-enmity and stable-alliance rates; *Pride and Prejudice* the Romance genre's elevated reconciliation rate; and *Great Expectations* the Coming-of-Age genre's dominant stable-alliance motif.

6. Genre Classification

6.1. Overall performance

Table 7.4 reports macro-averaged F_1 scores for the genre classification task (eight classes, 10-fold stratified cross-validation, $N = 726$ genre-labeled books, 142-dimensional feature vector).

Table 7.4.: Genre classification performance (macro-averaged F_1, 10-fold stratified CV, $N = 726$ genre-labeled works; 142-dimensional temporal and static feature vector).

Model	**Macro-F_1**
Majority-class dummy	0.071
Stratified-random dummy	0.127
Logistic Regression (L1-regularised)	0.196
Random Forest (500 trees, class-balanced)	0.139

The Random Forest achieves macro-$F_1 = 0.139$ and Logistic Regression 0.196, both substantially above the stratified-random baseline (0.127) and approximately twice the majority-class baseline (0.071). The full per-class precision/recall/F1 report is given in Appendix D, Table 4.2. Network structure therefore carries reliable genre signal at the corpus level. The modest absolute F_1 values reflect genuine structural overlap among genres rather than a pipeline limitation: the temporal network dynamics of a Coming-of-Age novel and a Romance are sufficiently similar that no feature set reliably separates them. This quantifies the degree to which network structure is genre-discriminative rather than genre-deterministic.

Figure 7.13 shows that Tragedy/Drama achieves the highest recall (0.31, $F_1 = 0.36$), driven by its structural distinctiveness in density and polarity. Literary/Realist Fiction acts as the dominant absorbing class (recall = 0.92): its structural profile overlaps all other novel-dominated genres, and it captures the plurality of misclassified books from every class. The novels-only restriction ($n = 537$, macro-$F_1 = 0.145$) confirms that Drama's form-driven distinctiveness provides only marginal uplift to the full-corpus macro-average, and that structural indistinguishability among novel genres is intrinsic rather than an artefact of including play-dominated classes.

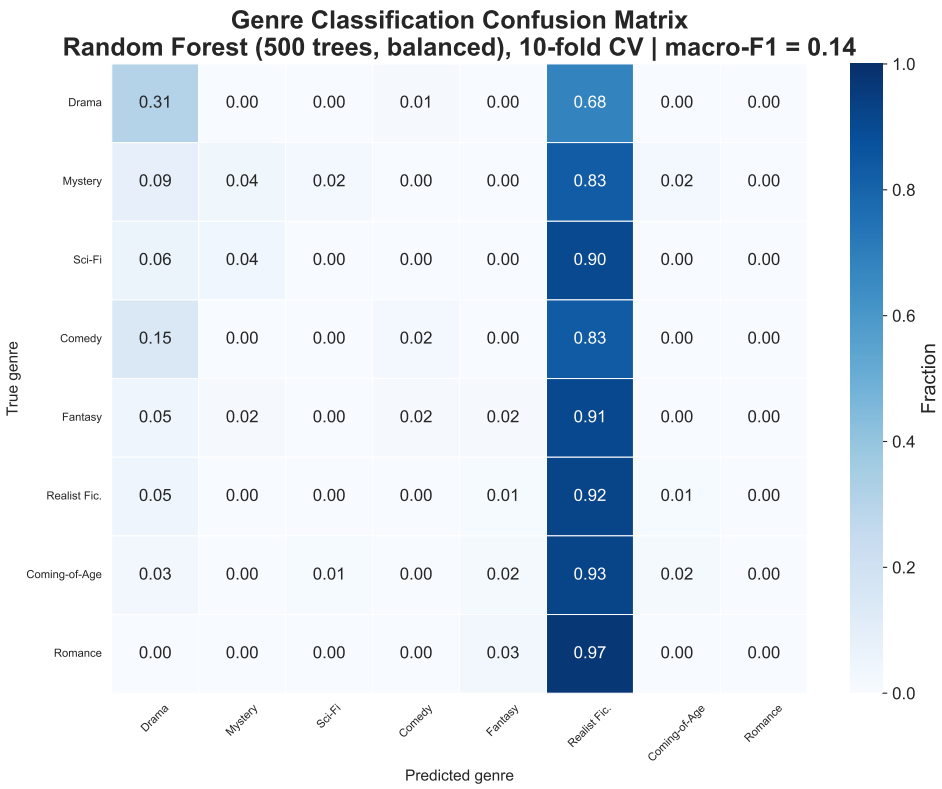

Figure 7.13.: Genre classification confusion matrix (Random Forest, 500 trees, balanced class weights, 10-fold CV; macro-$F_1 = 0.139$; $N = 726$). Row entries are row-normalised fractions. Tragedy/Drama achieves the highest recall of any class (0.31; $F_1 = 0.36$). Literary/Realist Fiction is the dominant absorbing class (recall = 0.92), capturing the plurality of misclassified books from all other novel-dominated genres.

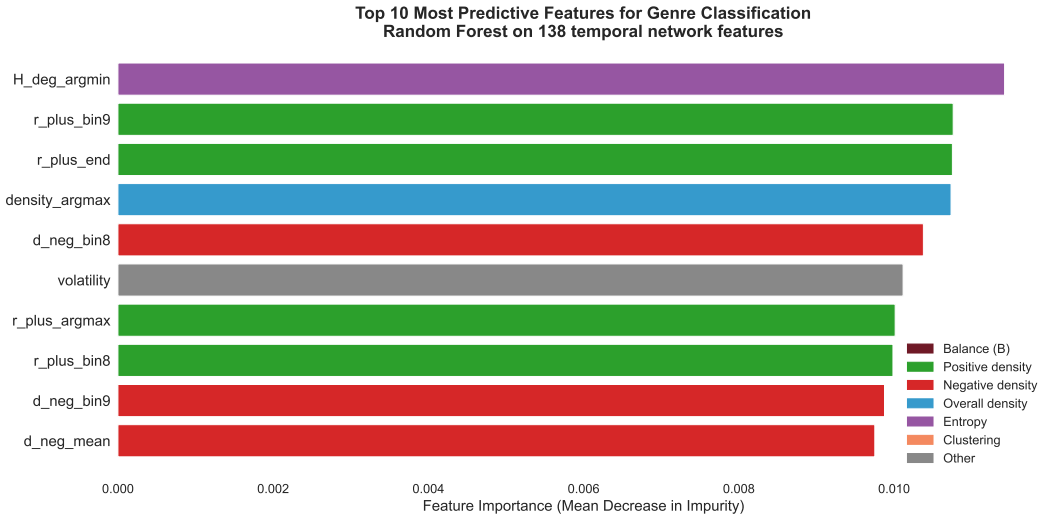

Figure 7.14.: Top 10 most predictive features for genre classification by mean decrease in impurity (Random Forest, 142-dimensional feature vector). The leading feature is the narrative position of minimum degree entropy (`H_deg_argmin`). Positive-density trajectory features (`r_plus_*`) account for four of the top ten positions; negative-density trajectory features (`d_neg_*`) account for three. No balance-index features appear in the top ten.

6.2. Feature importance

Figure 7.14 shows feature importance by mean decrease in impurity. The top-ranked predictor is the narrative position of minimum degree entropy (`H_deg_argmin`): the timing at which connectivity is most concentrated is the single most genre-discriminative signal, consistent with the genre-level entropy divergence in Figure 7.8. The next nine positions are dominated by positive-density trajectory features (`r_plus_*`: four positions) and negative-density trajectory features (`d_neg_*`: three positions). No $B(\tau)$ features appear in the top ten, consistent with the temporal flatness of the balance index documented in Section 4.

6.3. Pairwise discriminability

Binary Random Forest classifiers (5-fold CV) confirm the eight-class picture. Genre pairs involving Drama are the most separable: Drama vs Coming-of-Age reaches $F_1 = 0.802$, Drama vs Mystery and Drama vs Fantasy both reach $F_1 = 0.745$, and Drama vs Sci-Fi reaches $F_1 = 0.709$, all attributable to Drama's structural distinctiveness across density, polarity, and motif dimensions. Within strictly novel genres, Comedy/Satire vs Science Fiction/Dystopia is the most separable pair ($F_1 = 0.706$), followed by Comedy vs Romance and Sci-Fi vs Romance (both $F_1 = 0.704$ and 0.697 respectively). The least separable pairs are Coming-of-Age vs Romance ($F_1 = 0.426$) and Science Fiction/Dystopia vs Mystery ($F_1 = 0.498$); these genre pairs occupy nearly indistinguishable regions of the temporal network feature space. Most novel–novel pairs involving Realist Fiction fall below $F_1 = 0.55$, and no novel–novel pair exceeds $F_1 = 0.71$, confirming that novel genres form a common structural space that temporal network features alone cannot reliably partition.

The t-SNE projection (Figure 7.15) visualizes the structural geography of the feature space. Drama and Comedy form a partially separated peripheral cluster, reflecting their form-driven structural distinctiveness. The six novel-dominated genres occupy a diffuse, heavily overlapping central region with no clearly bounded clusters, confirming that genre boundaries in this feature space are gradients rather than discontinuities.

7. Historical Epoch Effects

Figure 7.16 plots the structural balance index $B(\tau)$ stratified by six literary epochs defined from publication year: Pre-Modern (pre-1600, $n = 38$), Early Modern (1600–1800, $n = 47$), 19th Century ($n = 109$), Early 20th Century

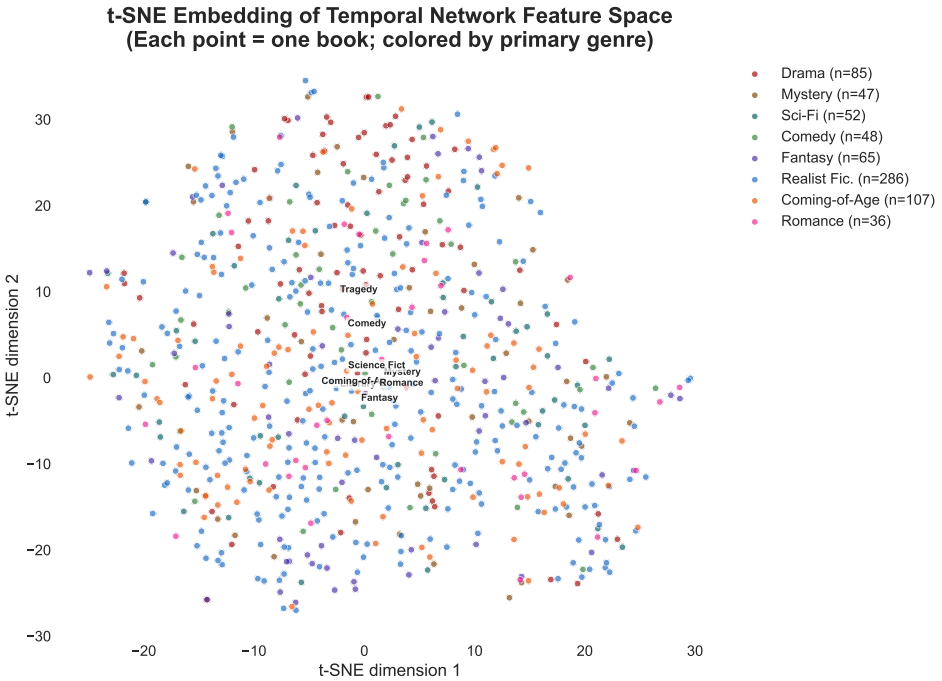

Figure 7.15.: t-SNE projection of the 142-dimensional temporal network feature space, coloured by genre. Tragedy/Drama and Comedy/Satire (play-dominated genres) form a partially separated peripheral cluster. All novel-dominated genres occupy a heavily overlapping central region with no clearly bounded class clusters, consistent with the limited pairwise discriminability among novel genres (Table 7.4).

(1900–1950, $n = 154$), Late 20th Century (1950–2000, $n = 298$), and 21st Century (2000+, $n = 203$).

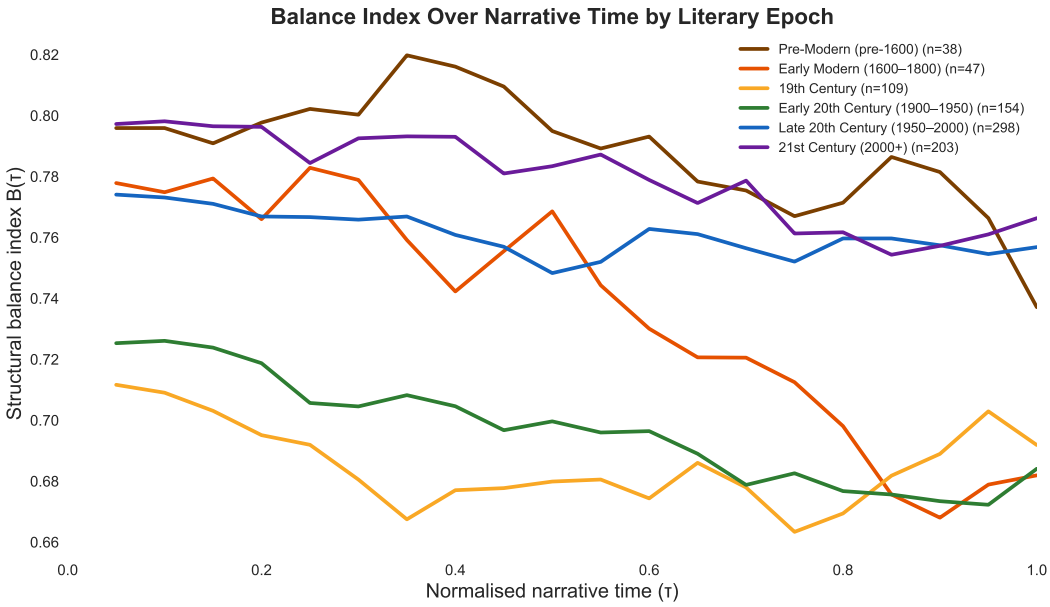

Figure 7.16.: Structural balance index $B(\tau)$ by literary epoch (mean trajectories). A non-monotonic pattern is visible: the Pre-Modern and 21st-Century cohorts occupy the highest balance band ($B \approx 0.79$–0.82); the 19th-Century and Early 20th-Century cohorts occupy the lowest ($B \approx 0.69$–0.73). The Early Modern cohort shows a pronounced narrative-time decline, falling from approximately $B = 0.78$ at $\tau = 0.1$ to $B = 0.68$ at $\tau = 1.0$. Late 20th Century occupies an intermediate position ($B \approx 0.75$–0.77).

Non-monotonic ordering. The epoch stratification reveals a non-monotonic pattern. The Pre-Modern cohort occupies the highest balance band ($B \approx 0.79$–0.82); the 21st-Century cohort sustains the second-highest position ($B \approx 0.79$–0.80); the 19th-Century and Early 20th-Century cohorts the lowest ($B \approx 0.69$–0.73), with Late 20th Century in an intermediate position ($B \approx 0.75$–0.77). The observed range between the highest and lowest epoch bands is approximately $\Delta B \approx 0.10$–0.12, of comparable order to the genre stratification effect documented in Section 4.1. Epoch therefore constitutes an independent source of structural variation in fictional social worlds.

A notable deviation from temporal flatness is visible in the Early Modern cohort: $B(\tau)$ declines from approximately 0.78 at $\tau = 0.1$ to 0.68 at $\tau = 1.0$, the steepest within-narrative decline of any epoch group. This trajectory

contrasts with the near-flat profiles of the novel-dominated 19th- and 20th-century cohorts, and may reflect the climactic concentration of antagonism in Renaissance and early Baroque dramatic plots — a form-content interaction that warrants closer examination in future work.

Interpretation. The U-shaped ordering is interpretable through the history of Western literary form. The 19th-Century cohort is dominated by realist novels, whose morally ambiguous characters, shifting loyalties, and unresolved social tensions generate relational configurations that do not conform to the clean alliance-and-enmity logic favored by strong-criterion structural balance [39, 53]. The Early 20th-Century cohort deepens this tendency: modernist fiction foregrounds fragmented consciousness, unreliable narration, and the dissolution of stable social categories, all of which produce more ambiguous, less balanced triad configurations. Pre-Modern works and contemporary popular fiction, at the two temporal extremes, appear to share a structural preference for legible social worlds with clearly delineated allies and antagonists.

Confounds. Two confounds limit direct interpretability. First, the epoch–form confound mirrors the genre–form confound discussed in Section 4.3: the Pre-Modern cohort is predominantly classical drama and Renaissance comedy (plays), while the 19th- and Early 20th-Century cohorts are predominantly realist novels. The lower balance of intermediate epochs may therefore partially reflect the form effect documented in Section 4.3 rather than a purely historical shift in narrative convention. Second, corpus composition is not epoch-balanced: Realist Fiction ($n = 286$) contributes the majority of 19th- and Early 20th-Century entries, concentrating any genre-specific balance depression in those cohorts. Disentangling epoch, genre, and form effects requires a form- and epoch-balanced corpus design, which is deferred to future work.

The epoch U-shape is not present in the 626-book conference corpus analysis, which restricted structural results to genre stratification. It constitutes an independent contribution of the extended corpus, offering a longitudinal perspective on how fictional social worlds reflect (or resist) the dominant literary-historical conventions of their period, and extending computational literary scholarship's evidence of systematic statistical shifts across the 19th and 20th centuries [53] to the domain of relational social structure.

Chapter VIII.

Discussion

This chapter situates the empirical findings reported in Chapter VII within the broader context of network science and literary theory, addresses structural properties that require interpretation beyond their face value, and examines the methodological limits of the present work.

1. What Temporal Networks Reveal About Fiction

Convergence with real-network laws. Two of the corpus' most robust findings — universal small-world structure ($\sigma_{\mathrm{sw}} > 1$ in 99.3 % of networks; corpus median $\tilde{\sigma}_{\mathrm{sw}} = 2.83$) and near-universal superlinear densification (97.1 % of books; median $\tilde{\alpha} = 1.55$) — mirror structural signatures of empirical social networks (Sections 1.2 and 2.2). The convergence is substantively interpretable: narrative construction favors compact, densely transitive social worlds because characters must remain legible to readers, and genre conventions modulate but do not override these regularities. Literary texts thus constitute a distinctive model system for social network science: intentionally constructed, culturally varied, yet structurally comparable to genuine social interaction.

The convergence is substantively interpretable. Narrative construction is not arbitrary: authors are constrained by the requirement that characters remain legible to readers, which favors compact, densely transitive social worlds over sprawling, weakly connected ones. Genre conventions further modulate but do not override these structural regularities: the small-world property holds across all eight genre classes despite substantial variation in cast size, edge polarity composition, and balance level (see Section 1). In this sense, literary texts constitute a distinctive model system for social network science: intentionally constructed, culturally varied, and structurally comparable to genuine social interaction at the network level.

Burstiness as the structural exception. The exception to this convergence is instructive. Inter-appearance burstiness κ is negative for every genre–polarity combination in the corpus ($\kappa \approx -0.67$ to -0.45; Section 3.2), indicating that character co-appearances are more regular than a Poisson process. This is precisely the opposite of bursty patterns — heavy-tailed inter-event intervals — observed in real human interaction networks [6]. The departure is not a failure of the literary network model but its most revealing property: unlike organic social dynamics, where temporal irregularity emerges from competing demands and contextual interruptions, authorial choices impose deliberate structural periodicity on character interactions. Co-appearances in fiction are timed; relationships are advanced or resolved at narratively appropriate moments. Negative burstiness thus marks the boundary between authored narrative and emergent social interaction, and is consistent with the parallel finding that edge volatility in this corpus (37.4%) is substantially lower than in real interaction networks of comparable density.

The combination of real-network structural laws with sub-Poisson temporal regularity suggests a productive framing for future work: literary networks are social networks with an authorial constraint imposed on their dynamics. Network science can characterize the structural envelope within which narrative construction operates; genre conventions and individual authorial choices determine where within that envelope a given work falls.

2. Structural Balance in Narrative Context

Why temporal flatness is a substantive finding. It might seem that if $B(\tau)$ is flat, the temporal approach adds nothing over a static cross-sectional measurement. This reasoning inverts the epistemic situation. A static analysis, computing B once per book from the cumulative graph, would yield a single scalar that could not distinguish between three qualitatively different trajectories: balance that is uniformly high throughout, balance that rises monotonically to a high endpoint, or balance that fluctuates but happens to average to a high value. (The temporal flatness itself is documented in Section 4.1; this section interprets its theoretical implications.) All three would be observationally equivalent under a static design. Temporal resolution is what allows us to *establish* that the trajectory is flat — that fictional social worlds reach and sustain high balance from early narrative positions rather than converging toward it progressively.

This finding bears directly on the continuous-time model of structural balance developed by [35], which predicts that signed network dynamics converge to

one of two absorbing states: global harmony (all positive ties) or two-faction separation (all between-group ties negative, all within-group ties positive). The persistently high and flat $B(\tau)$ trajectories across all eight genres are consistent with the absorbing-state interpretation: fictional networks appear to begin near the balanced absorbing state and remain there throughout the narrative. Importantly, this configuration is maintained not through universal alliance but through the dominance of common-enemy triads (+--), which emerge as the single most prevalent triad type across all genre classes (Section 5). Shared antagonism, not universal friendship, is the primary structural mechanism sustaining high balance in literary fiction. Coming-of-Age and Fantasy/Adventure display the highest balance levels precisely because their alliance-heavy polarity composition generates abundant common-enemy configurations among the protagonist's circle and the antagonist's associates; Tragedy/Drama's lower balance reflects a higher proportion of unresolved rivalries that produce genuinely imbalanced triads.

Where structural balance analysis adds insight. SBT provides three substantive contributions to the analysis of this corpus beyond what unsigned network metrics deliver.

First, the genre stratification of $B(\tau)$ (with Fantasy/Adventure and Coming-of-Age at the upper band and Tragedy/Drama at the lower — partially a form effect quantified in Section 3) is detectable only because signed relationships are available; unsigned density or clustering does not distinguish these genre-level patterns with the same reliability.

Second, the non-monotonic U-shaped epoch effect (Pre-Modern and 21st-Century works at the highest balance levels, Early 20th Century at the lowest) is exclusively a signed-network finding. The pattern aligns with the historical arc of psychological realism: the 19th- and early 20th-century cohorts coincide with the emergence of modernist fiction and its morally ambiguous character relationships, which plausibly produce more imbalanced triadic configurations than either classical drama or contemporary popular fiction [53, 39]. This epoch effect is of comparable magnitude to the genre effect ($\Delta B \approx 0.13$) and constitutes an independent structural dimension of the corpus that unsigned methods would not surface.

Third, the Freytag test, locating $\tau^* = \operatorname{argmin} B(\tau)$ and testing whether it falls in the canonical climax window $[0.60, 0.75]$, yields a global median $\tilde{\tau}^* = 0.667$ and a significantly non-uniform distribution (KS test, $p < 10^{-6}$), providing quantitative support for the hypothesis that narrative tension concentrates in the late-middle of fiction across genres.

Where structural balance does not discriminate. SBT is least discriminative among genre pairs whose networks occupy overlapping structural space, specifically among predominantly novel-dominated genres (Literary/Realist Fiction, Coming-of-Age, Fantasy/Adventure, Science Fiction, Mystery/Thriller, Romance). The pairwise discriminability matrix (Section 6) shows that most novel–novel pairs fall below $F_1 = 0.60$ in binary Random Forest classification.

Classification performance (RF macro-$F_1 = 0.139$; Section 6) is modest but above the stratified baseline, reflecting genuine structural overlap rather than pipeline failure: five concurrent factors — small network size, noisy genre labels, class imbalance, the temporal flatness of $B(\tau)$, and the form–genre confound — together predict low separability even when the pipeline operates correctly (see Section 4 for elaboration).

3. The Form–Genre Confound

The play–novel distinction is the most consequential structural confound in the corpus and must be treated explicitly in any comparative analysis involving Tragedy/Drama or Comedy/Satire. Tragedy/Drama is 92% plays ($n = 78$ of 85); Comedy/Satire is 69% plays ($n = 33$ of 48). All other genre classes are predominantly or exclusively novels (Chapter III). Plays and novels differ structurally in ways that are independent of genre content: dramatic texts organize action around a small, fixed cast that interacts repeatedly across a small number of scenes, whereas novels introduce characters progressively and sustain interaction across many more partitions. The result is that play-dominated genres exhibit systematically higher density, lower cast size, and lower small-world coefficients than equivalent novel-dominated genres, regardless of narrative content.

These structural differences have direct consequences for signed-network analysis. The lower balance of Tragedy/Drama in the full-corpus result, cited throughout Chapter VII as a genre-level signature, is substantially attributable to dramatic form rather than tragic narrative content. Form-stratified robustness checks (Figure 7.5) show that within the plays-only stratum, Tragedy and Comedy are nearly indistinguishable in balance ($B \approx 0.67$–0.72 with overlapping confidence intervals); the Tragedy–Comedy gap that appears in the full-corpus analysis collapses when form is held constant. Within the novels-only stratum ($n = 623$), the remaining six genre classes maintain their relative ordering and the Drama-associated balance depression disappears entirely.

What can be concluded is therefore limited to the following. Genre differences in structural balance, and in temporal network metrics generally, are reliable

where they obtain among form-matched groups: novel-to-novel genre comparisons can be interpreted as genre-content effects. The novel-level macro-F1 of 0.145 (vs. 0.139 for the full corpus) is a more appropriate benchmark for evaluating genre separability from network features, since it is uncontaminated by the form signal. What cannot be concluded is that Tragedy's lower balance reflects dramatic narrative content: that claim would require a corpus balanced across form within genre, with plays and novels represented equally within each genre class.

A form-balanced design, either stratifying the current corpus or augmenting it with novel-format tragic/comedic works, is deferred to future work.

4. Limitations

(i) Summary-derived corpus. All character networks are constructed from chapter-level narrative summaries, not primary texts. The resulting networks measure the structure of these summary representations rather than the full social fabric of the underlying texts. Summaries systematically underrepresent secondary characters; cast sizes and edge counts are likely lower bounds on full-text statistics. The pipeline is designed to transfer to full-text corpora in subsequent work.

(ii) LLM extraction limitations. LLM extraction at $T = 0.1$ is near-deterministic but not infallible. The three most likely failure categories are polarity inversion in indirect discourse, alias fragmentation for characters not in the supplied list, and underweighting of long-range dynamics beyond the three-partition history window. Benchmarked against human annotations (Chapter VI), the pipeline attains macro-$F_1 = 0.58$, within the inter-annotator agreement range ($\kappa \approx 0.65$). Detailed inclusion and exclusion criteria are listed in Appendix B.

(iii) Epoch–genre confound. The six literary epochs are not genre-balanced. Pre-Modern works (pre-1600) are predominantly classical drama; 19th-Century and Early 20th-Century works are predominantly realist novels; the Late 20th and 21st-Century cohorts contain substantial genre fiction. The U-shaped epoch effect in structural balance (Section 7) is therefore not cleanly separable from the genre-level variation documented in Section 4: the low balance of Early 20th Century could reflect the predominance of psychological realist novels in that cohort rather than, or in addition to, a purely historical effect. A design that balances genre composition across epochs would be required

to disentangle these contributions, which is not feasible in the present corpus without substantially reducing sample sizes per cell.

(iv) Class imbalance. The corpus is unevenly distributed across genre classes: Literary/Realist Fiction ($n = 286$) is 8× larger than Romance/Domestic Fiction ($n = 36$) and 6× larger than Comedy/Satire ($n = 48$). Although balanced class weights were applied in all Random Forest classifications (Section 6), class imbalance affects not only classifier training but also evaluation: macro-averaged F1 weights minority classes equally to majority classes, which inflates the apparent difficulty for classes with few support examples even when prediction in those classes is reasonable. Kruskal-Wallis tests throughout Chapter VII are non-parametric and do not require balanced classes, but post-hoc pairwise comparisons involving small-n genres (Romance, Comedy/Satire) should be interpreted with awareness of reduced power.

(v) Undefined balance estimates for sparse books. The minimum five-signed-triad threshold for $B(\tau)$ computation excluded 268 of 873 books (31%), predominantly short works with sparse signed interactions or small casts. This exclusion is not random: it disproportionately removes works with few characters, which tend to differ structurally from longer, more densely populated texts. Balance-based findings therefore characterize a subset of the corpus ($n_{\text{valid}} = 605$) whose structural properties may not generalize to the excluded fraction. The threshold of five triads was empirically calibrated as the minimum at which per-genre mean $B(\tau)$ trajectories stabilize (Chapter V); lowering it would admit noisier estimates without resolving the fundamental sparsity problem in small-cast works.

Chapter IX.

Conclusion

1. Summary of Contributions

This thesis has made four contributions to computational literary studies and temporal network science.

(i) An end-to-end extraction pipeline. An automated pipeline converts chapter-level narrative summaries and curated character lists into temporal signed character networks via LLM-based relation extraction (Meta Llama-3.3-70B-Instruct), assembling cumulative signed graphs $G_1, \ldots, G_T$ under a most-recent-wins convention. The pipeline is crash-resilient and designed to transfer without modification to full primary texts.

(ii) A corpus of 873 temporal signed character networks. The corpus spans eight genres and six historical epochs, comprising 105,550 character-relationship edges across 544 authors. Each work is represented by the full sequence of per-chapter edge tuples, the assembled cumulative graphs, and a 142-dimensional feature vector. To our knowledge, this is the largest dataset of temporal signed character networks released to date; materials are available at `https://github.com/L-Earthling/litnet-balance`.

(iii) Empirical analyses of structural and temporal regularities. Four converging empirical results are established across the corpus: fictional character networks are universally small-world and densify superlinearly (Sections 1 and 2); structural balance is persistently high and temporally flat, with genre stratification substantially attributable to a form–genre confound (Sections 4 and 3); temporal motifs confirm dominance of stable relationships with genre-specific variation in collapse and reconciliation rates (Section 5); and a non-monotonic U-shaped epoch effect in balance emerges as a co-equal source of structural variation alongside genre (Section 7).

(iv) Validation against human annotations. Benchmarked against majority-vote human annotations from Massey et al. [36], the pipeline attains macro-$F_1 = 0.58$ across 966 matched character pairs in 91 works, within the inter-annotator agreement range ($\kappa \approx 0.65$), establishing the first quantitative benchmark of LLM-based signed-relation extraction for literary texts (Chapter VI).

2. Answers to Research Questions

RQ1 (Structural regularities). Fictional character networks replicate the structural laws of real social networks with high regularity. Of 696 works with sufficient density, 99.3 % satisfy $\sigma_{\text{sw}} > 1$ (median $\tilde{\sigma}_{\text{sw}} = 2.83$), confirming universal small-world structure. A median densification exponent of $\tilde{\alpha} = 1.48$ with 97.1 % of works yielding $\alpha > 1$ confirms superlinear densification. These structural laws hold across all eight genre classes and constitute the most robust and interpretable findings of the thesis: narrative construction constrains authors to build social worlds that mirror the organizational properties of genuine social interaction.

RQ2 (Structural balance trajectories). The strong-balance index $B(\tau)$ is persistently high and temporally flat across all eight genres (all genre-mean trajectories in $[0.65, 0.84]$; no genre shows a systematic monotone trend). Genre stratification is statistically reliable ($H = 38.8$, $p = 2.1 \times 10^{-6}$, $\eta^2 = 0.062$) but substantially reflects the plays-versus-novels form confound for the two play-dominated genres. The global median of the per-book balance minimum falls at $\tilde{\tau}^* = 0.667$, within the Freytag climax window $[0.60, 0.75]$, and the distribution is significantly non-uniform ($p < 10^{-6}$), providing quantitative support for narrative tension concentrating in the late-middle portion of a work.

RQ3 (Temporal interaction motifs). Stable-alliance ($+++$) and stable-enmity ($---$) motifs together account for 76–87 % of all length-3 polarity sequences, confirming H3 (persistence dominance) in all genres. Comedy/Satire shows the highest rates of alliance-collapse and reconciliation, consistent with comedic arcs of misunderstanding and resolution. Tragedy/Drama is the only genre in which antagonism rivals alliance in base rate ($--- = 0.404$, $+++ = 0.389$).

RQ4 (Genre discriminability). Temporal network features carry a modest but statistically reliable genre signal: the Random Forest classifier achieves macro-$F_1 = 0.139$ against a stratified-random baseline of 0.127, with L1-penalized Logistic Regression reaching 0.196. The dominant predictive signal comes from entropy trajectory features rather than balance-index features — consistent

with the temporal flatness of $B(\tau)$, which limits its within-book variance and therefore its discriminative power. Tragedy/Drama is the best-recovered class ($F_1 = 0.36$); weak multi-class performance is interpreted as evidence of genuine structural overlap among literary genres rather than as a pipeline limitation.

Epoch finding (exploratory). Stratifying $B(\tau)$ by historical period reveals a non-monotonic U-shaped ordering: Pre-Modern and 21st-Century works occupy the highest balance band, while the 19th- and Early 20th-Century cohorts occupy the lowest. The effect magnitude ($\Delta B \approx 0.10$–0.12) is comparable to the genre effect and constitutes an independent structural contribution of the thesis. The pattern is interpretable through the historical arc of psychological realism and modernist fiction, though the epoch–genre confound limits clean causal attribution.

3. Limitations

The principal methodological limitations — summary-derived corpus, LLM extraction uncertainty, epoch–genre confound, class imbalance, and sparse triad exclusion — are documented in full in Section 4. Three limitations merit brief restatement in the context of this conclusion.

First, all networks are constructed from chapter-level summaries rather than primary texts. Cast sizes and edge counts are likely lower bounds; the structural findings are therefore conservative estimates of what full-text extraction would reveal.

Second, the form–genre confound prevents direct comparison of Tragedy/-Drama and Comedy/Satire against novel-dominated genres without form-stratified controls. Genre-level claims are reliable only within the novels-only stratum.

Third, the U-shaped epoch finding is exploratory rather than confirmatory: it was not predicted *a priori* and is entangled with genre composition across epochs. It is reported as a substantive observation warranting follow-up rather than a hypothesis-driven result.

4. Future Work

Three directions follow directly from the present work.

Extension to full primary texts. The most consequential next step is applying the same pipeline to full primary texts via Project Gutenberg. Full-text extraction will yield denser networks, better alias resolution via the full

narrative context, and coverage of secondary characters currently omitted from summaries. The pipeline is designed for this extension without structural modification.

Multilingual and cross-media comparison. Extending the corpus to non-English literary traditions and to other narrative media (film scripts, television series, graphic novels) would allow direct testing of whether the structural laws documented here — small-world structure, superlinear densification, high structural balance — are universal properties of authored narrative or are specific to the English literary canon.

Directed and weighted relationship extraction. The present pipeline extracts undirected, unweighted signed edges. Future work could pursue directed extraction (who initiates the relationship) and edge weighting by interaction frequency or confidence, enabling richer structural analyses including asymmetric balance measures and protagonist influence trajectories.

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Appendix A.

LLM Extraction Prompt Template

The following prompt template was used for all LLM-based extraction calls (see Section 3.2). Bracketed placeholders are substituted at runtime.

```
You are analyzing character relationships in "{
   book_name}" with a focus on PRECISION and CONTEXT.

CHARACTERS IN THIS STORY:
{character_context}

CURRENT PARTITION: {partition_title} (Chapter {
   partition_number})
PARTITION CONTENT:
{partition_content}
{history_context}

TASK: Extract character relationships considering:
1. Direct interactions in THIS partition
2. Relationship changes from previous chapters
3. Both explicit and implicit relationship indicators

RELATIONSHIP TYPES:
- POSITIVE: cooperation, friendship, support, alliance
   , shared goals
- NEGATIVE: conflict, hostility, opposition, betrayal,
    competing interests
- NEUTRAL: professional interaction, acquaintance,
   unclear sentiment

IMPORTANT RULES:
```

```
1. Include relationships even if unchanged from previous chapters
2. Consider subtle relationship indicators (tone, cooperation level)
3. Use canonical character names from the character list
4. If uncertain about relationship type, choose neutral
5. Include relationships for major characters even if interaction is brief

OUTPUT FORMAT: Return ONLY a JSON array of relationships:
[
  {
    "chapter": <partition_number>,
    "char_a": "Character Name",
    "char_b": "Character Name",
    "relationship": "positive|negative|neutral"
  }
]

If no clear relationships are found, return [].
```

Listing A.1: LLM extraction prompt template

When prior-partition history is available, the following block is prepended before the output format instruction:

```
PREVIOUS RELATIONSHIP CONTEXT (recent partitions):
  Ch{N-2}: CharA <-> CharB [positive]
  Ch{N-1}: CharC <-> CharD [negative]
  ...   (capped at 30 rows)
```

Listing A.2: History context block format

Appendix B.

Corpus Inclusion and Exclusion Criteria

Inclusion criteria

A work was included if it satisfied all of the following:

(i) English-language fiction (novels, plays, narrative poems with sustained character interaction).

(ii) Attributed to a single author and constituting a single self-contained work (not part of a multi-volume series edition).

(iii) Segmented into at least two narrative partitions in the source material.

(iv) The automated extraction pipeline produced at least one valid signed network with $N \geq 2$ distinct characters and $E \geq 1$ edges.

Exclusion criteria

A work was excluded if any of the following applied:

(i) Non-fiction: memoirs, autobiographies, essays, philosophical treatises, or histories.

(ii) Poetry without a sustained narrative character structure.

(iii) Multi-volume compiled editions, anthologies, or short-story collections presented as a single entry.

(iv) Networks with both large cast size ($N > 50$) and near-zero density ($\delta < 0.03$), indicating independent sub-stories with no shared characters (individually inspected before removal).

Special cases

Epic poems with sustained cross-partition character interaction, including *The Divine Comedy*, *The Aeneid*, and the *Mahabharata*, were *retained* despite their verse form, as they exhibit recurrent character interaction across partitions.

Works in translation were included if a canonical stable translation was available; the `Remarks` metadata field records the translation used.

Appendix C.

Case Studies: Per-Book Temporal Profiles

These case studies illustrate how the corpus-level findings reported in Chapter VII manifest at the level of individual works. Three books were selected to represent structurally distinct genres: *Hamlet* (Tragedy/Drama), *Pride and Prejudice* (Romance/Domestic Fiction), and *Great Expectations* (Coming-of-Age/Bildungsroman). Each subsection presents the per-book temporal profile figure generated by the analysis pipeline (Chapters IV and V), a full character relationship polarity heatmap, and a structural interpretation connecting the per-book trajectory to the genre-level findings in Chapter VII. All quantitative values are computed from the extracted networks and are consistent with the methodology described in Chapter V.

1. *Hamlet* (Shakespeare, 1623) — Tragedy/Drama

Hamlet is the work with which this thesis opens (Section 1), and its temporal character network exhibits precisely the structural trajectory that Section 1 describes qualitatively. The network is small ($N = 15$, $E = 32$ at $\tau = 1$, $\delta = 0.305$) and exhibits small-world structure ($\sigma_{\text{sw}} = 1.86 > 1$; Section 1.2). The final edge composition is $f^+ = 0.52$, $f^- = 0.36$, $f^0 = 0.11$, placing the play at the high end of the genre's negative-edge range and consistent with the Tragedy/Drama genre mean $\bar{f}^- = 0.405$ (Section 1.3).

The structural balance trajectory $B(\tau)$ is the most structurally distinctive feature of this work (Figure 3.1, top-center). Balance rises from $B = 0.75$ at the first valid slice ($\tau = 0.15$, four signed triads) to a plateau of $B = 1.0$ sustained from $\tau = 0.25$ to $\tau = 0.75$ — fifteen consecutive narrative partitions in which every closed signed triad is balanced. This opening phase of maximum balance

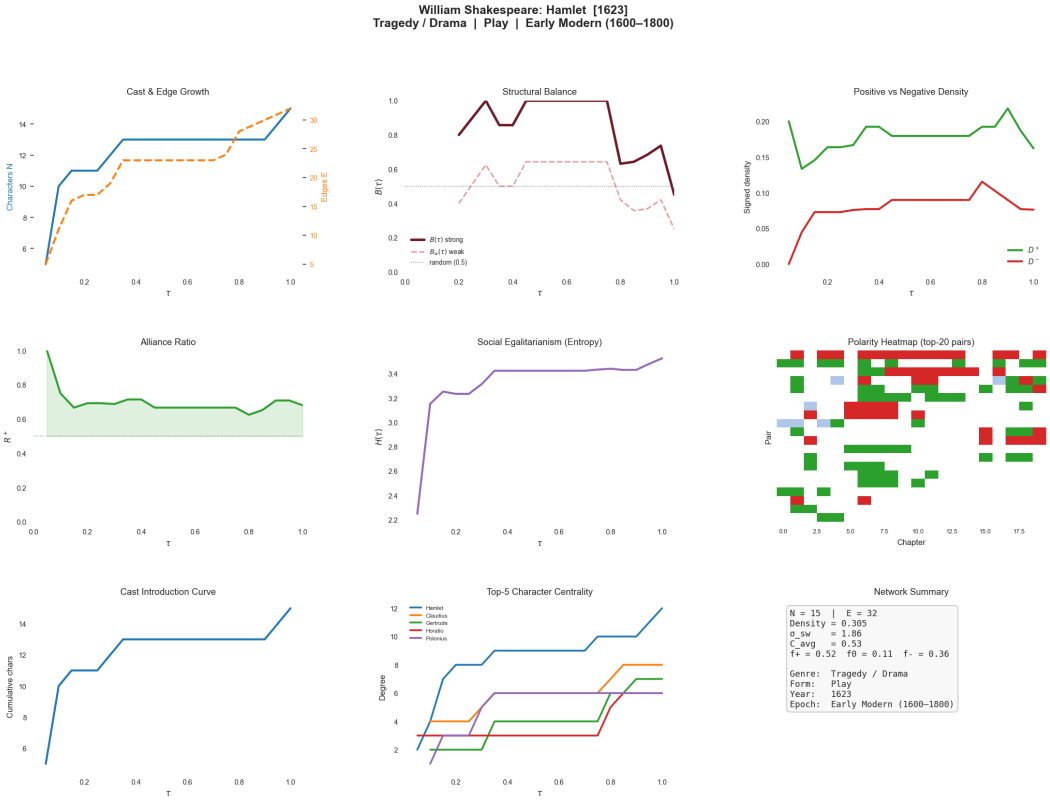

Figure 3.1.: Temporal network profile for *Hamlet* (Shakespeare, 1623; Tragedy/Drama; Play; $T = 20$ partitions, $N = 15$ characters, $E = 32$ cumulative edges at $\tau = 1$; $\sigma_{\text{sw}} = 1.86$). *Top row, left to right:* cast and edge growth over normalized narrative time τ; structural balance index $B(\tau)$ (strong criterion, dark line) and weak balance $B_w(\tau)$ (dashed) with the random-network expectation ($B = 0.5$, dotted); signed density trajectories $D^+(\tau)$ (positive, green) and $D^-(\tau)$ (negative, red). *Middle row:* alliance ratio $R^+(\tau)$; Shannon degree entropy $H(\tau)$; polarity heatmap for the top-20 most active character pairs (green: positive; red: negative; blue-grey: neutral; white: absent). *Bottom row:* cumulative cast introduction curve; degree centrality trajectories for the top-5 characters; network summary statistics.

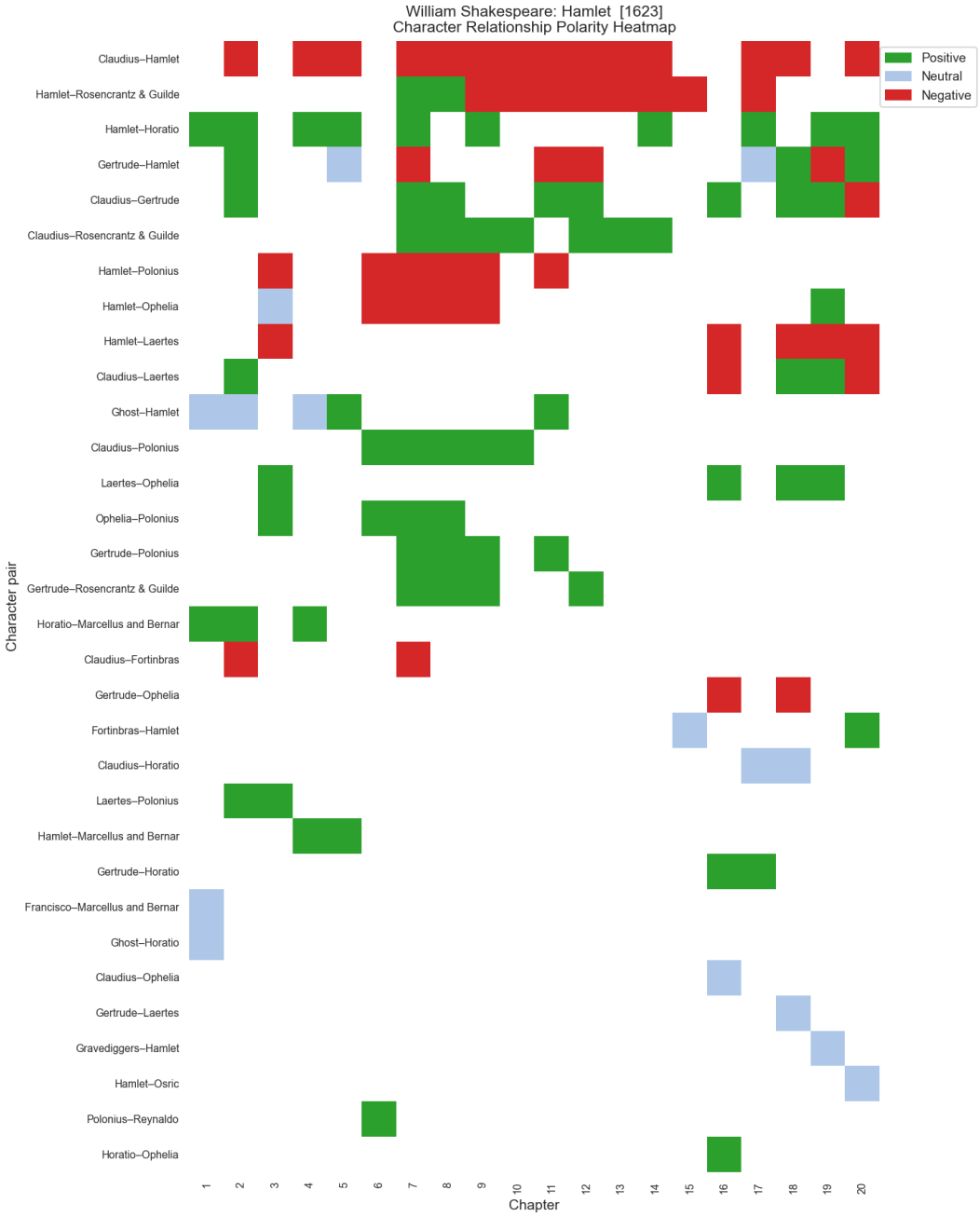

Figure 3.2.: Full character relationship polarity heatmap for *Hamlet*. Each row is a character pair; each column is a chapter (1–20). Colour encodes the cumulative edge polarity at that chapter under the most-recent-wins convention: green = positive, red = negative, blue-grey = neutral, white = pair not yet established. The Claudius–Hamlet row (top) is predominantly negative from Chapter 4 onward; the Gertrude–Hamlet row shows five sign changes; the Hamlet–Laertes row records a final positive reconciliation at Chapter 20.

reflects the initial alignment of Hamlet, Horatio and the Ghost against Claudius and Polonius, forming a stable two-faction configuration with no imbalanced triads despite mounting antagonism.

The structural discontinuity occurs at $\tau = 0.80$ (Act IV). B drops abruptly from 1.000 to 0.632 as five new signed edges enter the network simultaneously: the Claudius–Laertes alliance (positive) and the Claudius–Fortinbras antagonism (negative) introduce new triads that do not conform to the existing faction structure. The final chapter ($\tau = 1.0$) records the minimum balance of the entire trajectory at $B = 0.450$, concentrating five simultaneous sign flips: Claudius–Gertrude (positive $\rightarrow$ negative, as Gertrude drinks the poisoned cup), Claudius–Laertes (positive $\rightarrow$ negative at betrayal), Hamlet–Laertes (negative $\rightarrow$ positive at deathbed reconciliation), Gertrude–Hamlet (negative $\rightarrow$ positive), and Fortinbras–Hamlet (neutral $\rightarrow$ positive at the handoff of the state). The polarity heatmap (Figure 3.2) shows these final-chapter transitions clearly as simultaneous colour changes in the top rows.

The per-book argmin $\tau^* = 1.0$ — peak social tension at the very last chapter — is a structural outlier relative to the corpus-wide median $\tilde{\tau}^* = 0.667$ documented in Section 4.4. It reflects the Shakespearean tragic convention that social dissolution is irreversible and concentrated in the denouement, not the rising-action climax. This is also why Tragedy/Drama is the only genre in which stable-enmity motifs ($--- = 0.404$) rival stable-alliance motifs ($+++ = 0.389$) at corpus level (Section 5): Hamlet exemplifies the genre pattern in which antagonism, once established, persists without resolution.

2. *Pride and Prejudice* (Austen, 1813) — Romance/Domestic Fiction

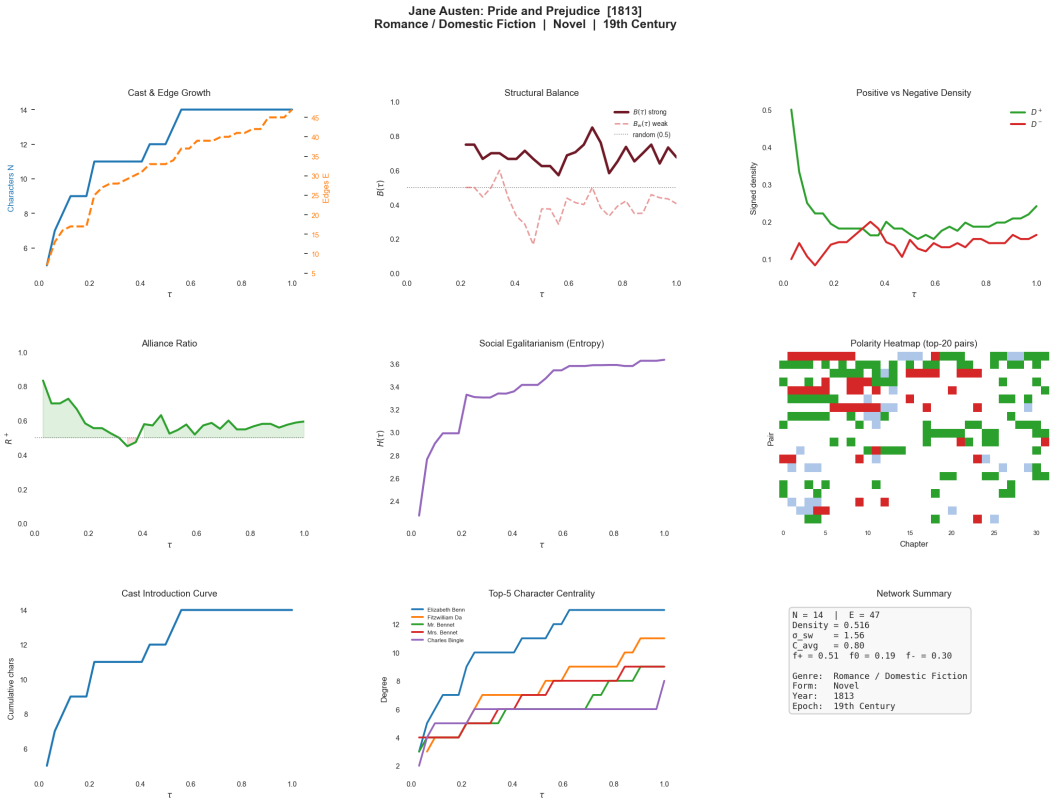

Figure 3.3.: Temporal network profile for *Pride and Prejudice* (Austen, 1813; Romance/Domestic Fiction; Novel; $T = 32$ partitions, $N = 14$ characters, $E = 47$ cumulative edges at $\tau = 1$; $\sigma_{\text{sw}} = 1.56$). Panel layout as in Figure 3.1. The alliance ratio panel (middle-left) shows $R^+(\tau)$ persistently above 0.5 for most of the narrative with a minimum around $\tau = 0.15$. The structural balance panel (top-center) exhibits high but volatile $B(\tau)$, oscillating in the range $[0.500, 0.850]$ with no systematic trend — the characteristic pattern of Romance-genre balance documented in Section 4.1.

Pride and Prejudice exemplifies the structural profile of the Romance/Domestic Fiction genre: the highest corpus-level positive-edge fraction ($\bar{f}^+ = 0.543$; Section 1.3), a densely connected cast ($N = 14$, $E = 47$, $\delta = 0.516$, $C_{\text{avg}} = 0.795$), and a narrative arc in which initial antagonisms progressively resolve. The final polarity composition is $f^+ = 0.51$, $f^- = 0.30$, $f^0 = 0.19$, with

Figure 3.4.: Full character relationship polarity heatmap for *Pride and Prejudice* (Austen, 1813; 32 chapters; 14 characters). The Elizabeth Bennet–Fitzwilliam Darcy row (top) records four sign changes across the narrative — negative → neutral → negative → positive — constituting the central reconciliation arc of the novel. The Elizabeth Bennet–George Wickham row shows a parallel disruption arc (positive → negative at mid-narrative revelation). Most other pairs remain in a single polarity throughout, producing the high aggregate balance visible in Figure 3.3.

the modest negative fraction reflecting the genre-level finding that Romance networks differentiate from other genres primarily through elevated positive rather than suppressed negative ties.

The structural balance index oscillates in a narrow band $B(\tau) \in [0.500, 0.850]$ throughout the narrative, with no monotone trend (Figure 3.3, top-center). This volatility is itself a finding: unlike *Hamlet*, where balance is stable for two-thirds of the narrative before collapsing, the social reorganization in *Pride and Prejudice* is continuous and episodic. The per-book minimum $B = 0.500$ occurs at $\tau^* = 0.156$ (Chapter 5), placing peak structural tension in the early narrative — consistent with the Romance genre median $\tilde{\tau}^* = 0.654$ (Section 4.4), and reflecting the early establishment of the Darcy–Elizabeth antagonism that the rest of the novel works to resolve.

The central structural trajectory is the Elizabeth Bennet–Fitzwilliam Darcy edge, visible as the topmost row in Figure 3.4. The pair cycles through four sign states: negative (Chapters 1–10), neutral (11–18), negative again (19), and finally positive from Chapter 20 onward. This four-state oscillation is the narrative's primary relational engine, and each transition corresponds to a plot-level revelation: Darcy's first proposal (Chapter 11, negative $\rightarrow$ neutral), his letter of explanation (Chapter 12, clarifying circumstances), Elizabeth's recognition of Wickham's deception (Chapter 16–19, recalibrating Darcy to negative), and her change of heart at Pemberley (Chapter 20–21, negative $\rightarrow$ positive). The parallel Wickham–Lydia trajectory (positive $\rightarrow$ negative $\rightarrow$ positive) and the Lady Catherine de Bourgh–Darcy inversion (positive $\rightarrow$ negative at the confrontation scene) add subsidiary sign-change sequences that are legible in the heatmap.

Pride and Prejudice thus illustrates the key genre-level finding that Romance balance is high and flat in aggregate while remaining dynamically active at the dyadic level (Section 4.1): the stable triad census masks a continuous reorganization of individual relationships that the motif analysis captures as an elevated reconciliation rate ($--+$ motif; Section 5).

3. *Great Expectations* (Dickens, 1861) — Coming-of-Age/Bildungsroman

Great Expectations provides the longest temporal resolution of the three case studies ($T = 59$ chapters) and the most structurally explicit Bildungsroman arc in the corpus. The network is modest in scale ($N = 19$, $E = 50$, $\delta = 0.292$) but records the highest degree entropy growth of the three works ($H(\tau) : 2.0 \to 4.0$; Figure 3.5, middle-center), reflecting the progressive diversification of Pip's social world as he moves from the marshes to London and back. The small-world coefficient $\sigma_{\text{sw}} = 1.90$ is the highest of the three case studies, consistent with the tight clustering of Pip's household network combined with the longer paths connecting to the London and Satis House sub-communities.

The structural balance trajectory reveals a U-shaped arc that is the network-level signature of the Bildungsroman genre (Figure 3.5, top-center; Section 4.1). Balance begins high (B=0.833 at $\tau = 0.14$, eight chapters in), then drops sharply to its global minimum of $B = 0.400$ at $\tau^* = 0.22$ (Chapter 13), corresponding to the period of Pip's maximum social dislocation: his estrangement from Joe Gargery (positive $\to$ negative, Chapter 12), his emerging antagonism with Mrs. Joe, and the contradictory claims of Magwitch's and Miss Havisham's influences. This early minimum ($\tilde{\tau}^* = 0.22$ for this work vs. the Coming-of-Age genre median $\tilde{\tau}^* = 0.573$; Section 4.4) reflects that narrative disruption in Bildungsromane arrives early, before the protagonist has constructed a stable adult social world.

From $\tau = 0.27$ onward, balance recovers monotonically. By $\tau = 0.30$ the network reorganizes into a London-centered alliance cluster (Herbert Pocket, Wemmick, and Pip), and balance stabilizes at $B \approx 0.50$–0.70 for the middle narrative. The final rise to $B = 0.833$ at $\tau = 1.0$ is achieved through three simultaneous resolutions: Magwitch's identity accepted (Abel Magwitch–Pip positive), Miss Havisham's confession (Miss Havisham–Pip positive after the fire), and the Estella–Pip relationship's final positive-polarity assignment after Estella's liberation from Bentley Drummle. The polarity heatmap (Figure 3.6) shows this as a progressive greening of the upper rows toward $\tau = 1$.

The alliance ratio $R^+(\tau)$ (Figure 3.5, middle-left) begins below 0.5 in the first five chapters — the only case among the three studies in which antagonism initially outweighs alliance — and rises above 0.5 permanently after $\tau \approx 0.20$, mirroring the balance recovery. Together, *Great Expectations* illustrates why the Coming-of-Age genre has the highest stable-alliance motif rate (+++ = 0.672; Section 5) despite beginning in social conflict: the protagonist's social integra-

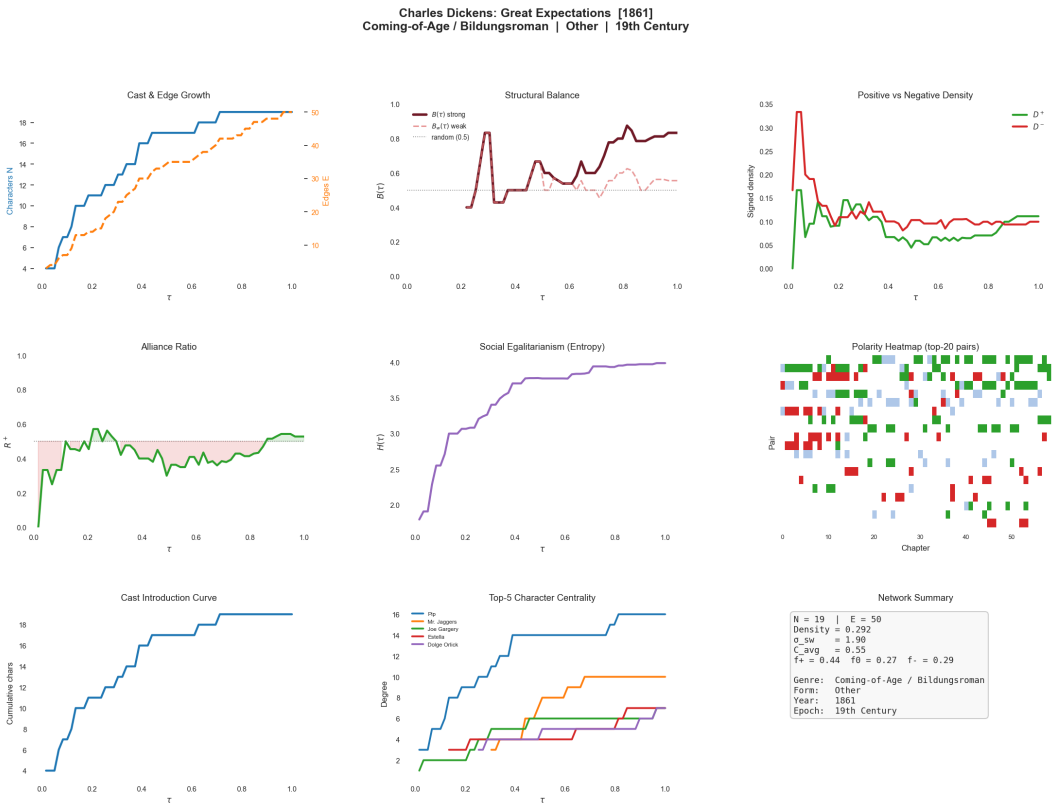

Figure 3.5.: Temporal network profile for *Great Expectations* (Dickens, 1861; Coming-of-Age/Bildungsroman; Novel; $T = 59$ partitions, $N = 19$ characters, $E = 50$ cumulative edges at $\tau = 1$; $\sigma_{\text{sw}} = 1.90$). Panel layout as in Figure 3.1. The structural balance panel (top-center) shows a sharp minimum at $\tau^* = 0.22$ followed by recovery to $B \approx 0.83$ at $\tau = 1$. The alliance ratio panel (middle-left) shows R^+ beginning below the 0.5 threshold (antagonism-dominated opening) and rising above it after $\tau \approx 0.20$. The degree entropy panel (middle-center) records the highest entropy growth ($H : 2.0 \to 4.0$) of the three case studies, consistent with the richest cast expansion and Pip's progressive social integration.

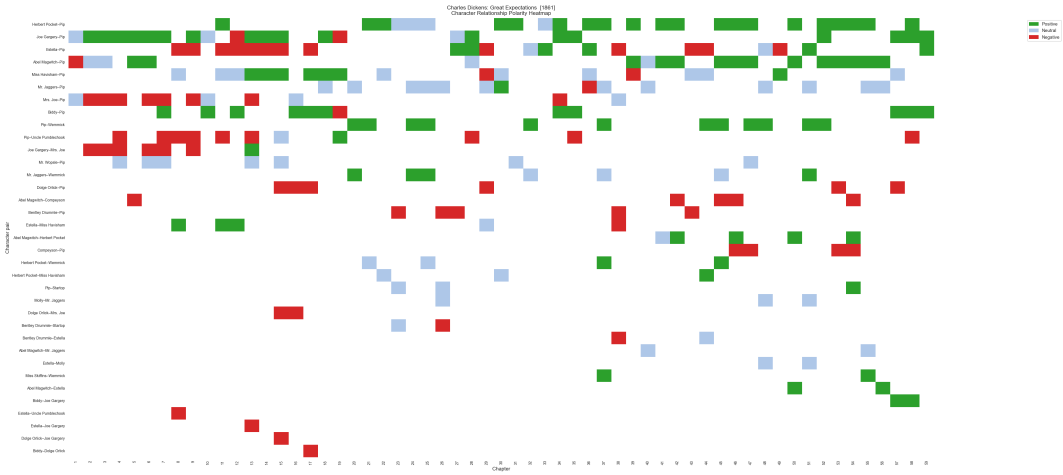

Figure 3.6.: Full character relationship polarity heatmap for *Great Expectations* (Dickens, 1861; 59 chapters; 19 characters). The heatmap is centered on Pip's bilateral relationships. The Estella–Pip row (third) shows the most volatile trajectory of the three case studies: five sign changes across 59 chapters. The Joe Gargery–Pip and Abel Magwitch–Pip rows both record initial conflict followed by eventual positive resolution. The Mrs. Joe–Pip row is predominantly negative throughout, providing the antagonism that anchors the early structural imbalance. The dense green region in the Herbert Pocket–Pip row reflects the stable positive friendship that persists from Pip's arrival in London to the narrative's close.

tion, compressed here across 59 chapters, converts an antagonism-dominated opening into a persistently positive terminal state.

Appendix D.

Additional Figures and Tables

Per-genre summary statistics

Table 4.1.: Per-genre summary statistics from the 873-book analysis corpus. $\tilde{N}$: median cast size; $\tilde{E}$: median edges; $\bar{\delta}$: mean density; $\bar{B}$: mean balance index (books with ≥ 5 signed triads per slice); $\bar{\kappa}$: mean inter-appearance burstiness (positive edges).

Genre	$\tilde{N}$	$\tilde{E}$	$\bar{\delta}$	$\bar{B}$	$\bar{\kappa}$
Coming-of-Age / Bildungsroman	15	27	0.283	0.779	−0.52
Science Fiction / Dystopia	15	27	0.314	0.769	−0.53
Fantasy / Adventure	20	38	0.274	0.762	−0.53
Literary / Realist Fiction	16	28	0.271	0.717	−0.58
Romance / Domestic Fiction	16	29	0.285	0.704	−0.45
Mystery / Thriller	14	25	0.365	0.684	−0.46
Comedy / Satire	14	25	0.369	0.668	−0.56
Tragedy / Drama	11	20	0.437	0.668	−0.66

Fill in $\bar{B}$ values from the genre-mean balance trajectory outputs.

Full genre classification report

Table 4.2.: Full per-class classification report, Random Forest (500 trees, balanced class weights, 10-fold CV; $N = 726$).

Genre	**Precision**	**Recall**	F_1	**Support**
Comedy / Satire	0.33	0.02	0.04	48
Coming-of-Age / Bildungsroman	0.33	0.02	0.04	107
Fantasy / Adventure	0.14	0.02	0.03	65
Literary / Realist Fiction	0.41	0.92	0.57	286
Mystery / Thriller	0.33	0.04	0.08	47
Romance / Domestic Fiction	0.00	0.00	0.00	36
Science Fiction / Dystopia	0.00	0.00	0.00	52
Tragedy / Drama	0.44	0.31	0.36	85
Macro avg	0.25	0.17	0.14	726
Weighted avg	0.32	0.41	0.28	726

Pairwise genre discriminability

Table 4.3.: Pairwise genre discriminability: binary Random Forest macro-F_1 (5-fold CV). Values above 0.60 are bolded.

	Drama	*Mystery*	*Sci-Fi*	*Comedy*	*Fantasy*	*Realist*	*Coming-Age*	*Romance*
Drama	—	**0.745**	**0.709**	0.581	**0.745**	**0.659**	**0.802**	**0.748**
Mystery		—	0.498	**0.684**	0.549	0.481	0.595	0.637
Sci-Fi			—	**0.706**	0.554	0.458	0.526	**0.697**
Comedy				—	**0.697**	0.460	0.591	**0.704**
Fantasy					—	0.543	0.618	0.580
Realist						—	0.471	0.470
Coming-Age							—	0.426

Appendix E.

Ethics and Licensing

Source material. The corpus is derived from publicly available chapter-level narrative summaries and character lists for literary works, collected from online study guide platforms. No personal data was collected or processed at any stage.

Derived dataset. The derived character network dataset (character nodes, edge tuples, polarity labels) and associated metadata are released under a Creative Commons Attribution 4.0 International (CC BY 4.0) license. Attribution to this thesis is required for reuse.

Code. All pipeline and analysis code is released under the MIT License.

LLM usage. Meta Llama-3.3-70B-Instruct was used exclusively for automated data extraction (relationship labeling per narrative partition). It was not used for any part of the writing of this thesis. Model weights are available under the Meta Llama Community License.

Appendix F.

API Infrastructure

Model. Meta Llama-3.3-70B-Instruct (`meta-llama/Llama-3.3-70B-Instruct`).

Inference endpoints.

- AcademicCloud Chat-AI (`chat-ai.academiccloud.de/v1`), rate-limited at 15 req/min, 900 req/hour per key.
- DeepInfra (`api.deepinfra.com/v1/openai`), no hard per-minute ceiling.

Processing configuration. Temperature = 0.1; maximum output tokens = 1,500; up to 3 retries per failed call; 6 API keys in round-robin rotation.

Wall-clock time. Approximately 132 hours total processing time on consumer hardware (Apple MacBook Pro, M-series chip). No GPU or specialized HPC infrastructure was required.

Reproducibility. All configuration parameters (model version, temperature, prompt text, rate-limit settings) are stored in a single configuration cell written to a run log at execution start. A SQLite progress database records completion status at partition granularity, enabling deterministic resumption after interruption.

Appendix G.

Environmental Impact Statement

LLM inference was performed on third-party cloud infrastructure (AcademicCloud and DeepInfra); exact energy consumption figures are not disclosed by these providers.

All analysis was performed on a laptop CPU; its footprint is negligible. No model training was performed.